FINAL Unit1 MCQ - Topic order - Deduplicated

1. A student wants to create an algorithm to calculate the combined total area of two rectangles. Which of the following algorithms can be used to perform this calculation?

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(A)

Step 1:	Multiply the length of the first rectangle by the length of the second rectangle.
Step 2:	Multiply the width of the first rectangle by the width of the second rectangle.
Step 3:	Add the result of step 1 to the result of step 2.

(B) ✓

Step 1:	Multiply the length of the first rectangle by the width of the first rectangle.
Step 2:	Multiply the length of the second rectangle by the width of the second rectangle.
Step 3:	Add the result of step 1 to the result of step 2.

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(C)

Step 1:	Add the length of the first rectangle to the width of the first rectangle.
Step 2:	Add the length of the second rectangle to the width of the second rectangle.
Step 3:	Multiply the result of step 1 by the result of step 2.

(D)

Step 1:	Add the length of the first rectangle to the length of the second rectangle.
Step 2:	Add the width of the first rectangle to the width of the second rectangle.
Step 3:	Multiply the result of step 1 by the result of step 2.

Answer B

Correct. Step 1 calculates the area of the first rectangle by multiplying its length by its width. Step 2 calculates the area of the second rectangle by multiplying its length by its width. Step 3 calculates the combined area by adding the areas determined in steps 1 and 2.

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2. A carpenter would like to know how many 8-inch-long pieces can be cut from a board whose length is measured in feet. Which of the following algorithms can be used to determine the number of pieces?

(A)

Step 1:	Divide the length of the board by 8.
Step 2:	Keep the whole number portion of the result of step 1, and discard the decimal portion.
Step 3:	Multiply the result of step 2 by 12.

(B)

Step 1:	Divide the length of the board by 12.
Step 2:	Keep the whole number portion of the result of step 1, and discard the decimal portion.
Step 3:	Multiply the result of step 2 by 8.

(C)

Step 1:	Multiply the length of the board by 8.
Step 2:	Divide the result of step 1 by 12.
Step 3:	Keep the whole number portion of the result of step 2, and discard the decimal portion.

(D) ✓

Step 1:	Multiply the length of the board by 12.
Step 2:	Divide the result of step 1 by 8.
Step 3:	Keep the whole number portion of the result of step 2, and discard the decimal portion.

Answer D

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Correct. The board length is first converted from feet to inches by multiplying by 12. That result is divided by 8 to calculate how many pieces can be cut. The whole number part represents the number of pieces. The decimal portion can be discarded.

3. A student wrote a program to calculate the average of several numeric scores. The program compiles without error but produces incorrect results when it runs. The student suspects there is a logic error in the code.

Which of the following strategies is most likely to help identify the logic error?

(A) Changing random parts of the code and rerunning the program until the error is resolved

(B) Inserting print statements throughout the code to check the calculations at various points in the execution ✓

(C) Relying on the compiler's error messages to detect the logic error

(D) Running the program multiple times without making changes to see if the error corrects itself

Answer B

Correct. Inserting print statements in code can show what the code is doing as it executes. This can reveal the point when an error or incorrect result is introduced.

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4. A student wants to create an algorithm to calculate the average (mean) of two quiz scores. Which of the following algorithms can be used to determine the correct average?

(A)

Step one:	Add the two quiz scores.
Step two:	Multiply the result of step one by 2 to determine the average.

(B) ✓

Step one:	Add the two quiz scores.
Step two:	Divide the result of step one by 2 to determine the average.

(C)

Step one:	Subtract the lower quiz score from the higher quiz score.
Step two:	Multiply the result of step one by 2 to determine the average.

(D)

Step one:	Subtract the lower quiz score from the higher quiz score.
Step two:	Divide the result of step one by 2 to determine the average.

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Answer B

Correct. An average is the sum of the list of numbers divided by how many numbers are in the list. In this case, there are two numbers, so the sum is divided by 2.

5. A student is writing a program that is intended to convert feet to meters. The student enters the code for the program into an integrated development environment (IDE), but the code fails to compile.

Which of the following steps is most likely to help the student resolve the errors that prevent the code from compiling?

(A) Adding print statements to find and fix any logic errors

(B) Rearranging the order of the statements in the code to fix any sequencing errors

(C) Removing an exception error message by using a different compiler

(D) Using the compiler's error messages to find and fix any syntax errors ✓

Answer D

Correct. A syntax error occurs when the rules of the programming language are not followed. These errors are detected by a compiler. The compiler will provide error messages to indicate where syntax errors are located in the code so that they can be corrected.

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6. A teacher wants to calculate the final grade for a student. The final grade is the average of three test scores that have equal weight and one final exam that is weighted twice as much as each test score. The tests and final exam are each scored on a scale of 0 to 100 points.

Which of the following algorithms can be used to determine the correct final grade?

(A)

Step 1:	Compute the sum of the three test scores and the final exam score.
Step 2:	Multiply the result of the step 1 by 2.
Step 3:	Divide the result of step 2 by 5.

(B) ✓

Step 1:	Compute the sum of the three test scores.
Step 2:	Multiply the final exam score by 2 and add it to the result of step 1.
Step 3:	Divide the result of step 2 by 5.

(C)

Step 1:	Compute the sum of the three test scores.
Step 2:	Divide the result of step 1 by 3.
Step 3:	Multiply the final exam score by 2.
Step 4:	Add the results of step 2 and step 3.

(D)

Step 1:	Compute the sum of the three test scores.
Step 2:	Divide the result of step 1 by 2.
Step 3:	Compute the sum of the result of step 2 and the final exam score.
Step 4:	Divide the result of step 3 by 2.

Answer B

Correct. To compute a weighted average, multiply each value by its weight, sum the products, and divide the sum by the sum of the weights. For each test, the weight is 1 and for the final exam, the weight is 2. The weighted average is

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calculated by finding the sum of the three test scores and 2 times the final test score, then dividing that sum by 5.

7. A code segment will use variables to represent the age of an animal and whether it is considered a house pet. Which of the following variable declarations is most appropriate for the code segment?

(A)

```
int age;

int pet;
```

(B) ✓

```
int age;

boolean pet;
```

(C)

```
double age;

double pet;
```

(D)

```
boolean age;

boolean pet;
```

Answer B

Correct. Ages are typically represented using non-decimal numbers, so an `int` variable is most appropriate. Boolean variables represent something that can be expressed as being `true` or `false`, such as classifying something as being a house pet.

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8. Consider the following code segment.

```
int x = 5;
int y = 6;
/* missing code */
z = (x + y) / 2;
```

Which of the following can be used to replace /* missing code */ so that the code segment will compile?

(A)

```
boolean z = true;
```

(B)

```
int z = 0.0;
```

(C) ✓

```
int z = 0;
```

(D)

```
z = 0;
```

Answer C

Correct. Since the variables x and y are declared as int data types, the result of the addition operation performed on those variables will be of type int. If z is declared as an int, the code will compile because z is assigned an int value.

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9. A code segment (not shown) is intended to determine the number of players whose average score in a game exceeds `0.5`. A player's average score is stored in `avgScore`, and the number of players who meet the criterion is stored in the variable `count`.

Which of the following pairs of declarations is most appropriate for the code segment described?

(A)
```
double avgScore;
boolean count;
```

(B) ✓
```
double avgScore;
int count;
```

(C)
```
int avgScore;
boolean count;
```

(D)
```
int avgScore;
int count;
```

Answer B

Correct. The criterion "exceeds `0.5`" indicates that the average score will be a fractional value; thus, the `avgScore` variable should be declared as a `double`. Since the `count` variable is intended to store a count of players, which is a nonnegative integer, it should be declared as an `int`.

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10. A code segment will use variables to represent a student's age, in years, and whether the student has a driver's license. Which of the following variable declarations are most appropriate for the code segment?

(A)

```
boolean age;

double hasLicense;
```

(B)

```
boolean age;

boolean hasLicense;
```

(C) ✓

```
int age;

boolean hasLicense;
```

(D)

```
int age;

double hasLicense;
```

Answer C

Correct. Ages are typically represented using non-decimal numbers, so an `int` variable is most appropriate. For whether or not something is true, like whether a student has a driver's license, a `boolean` variable is most appropriate.

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11. A program needs to maintain the following information about data captured by a sensor.

- Whether the sensor is currently active
- The number of items detected by the sensor
- The average time per detection, in seconds

Which set of variable declarations is most appropriate for this program?

(A)

```
int active;

double items;

boolean avgTime;
```

(B)

```
int active;

int items;

double avgTime;
```

(C)

```
boolean active;

int items;

boolean avgTime;
```

(D) ✓

```
boolean active;

int items;

double avgTime;
```

Answer D

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Correct. A `boolean` variable is most appropriate for indicating whether a sensor is active (`true` or `false`). An `int` variable is most appropriate for indicating the number of items, since that value is expected to be a whole number. A `double` variable is most appropriate for storing the average time, since that value is likely a decimal.

12. Consider the following variable declaration.

```
boolean b;
```

Which of the following values can be stored in the variable b?

(A)

-1

(B)

0

(C)

1.0

(D) ✓

false

Answer D

Correct. A boolean variable can be assigned only the values true or false.

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13. Consider the following variable declaration.

```
int x;
```

Which of the following values can be stored in the variable x?

(A)

0.5

(B) ✓

1

(C)

true

(D)

false

Answer B

Correct. An int variable can be assigned an integer value such as 1.

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14. Consider the following code segment.

```
int a = 5;
int b = 8;
int c = 3;
System.out.println(a + b / c * 2);
```

What is printed as a result of executing the code segment?

(A)

2

(B)

6

(C)

8

(D) ✓

9

Answer D

Correct. The integer division b / c is performed first, producing 2. Then 2 is multiplied by 2, producing 4. Finally, 4 is added to a, producing 9.

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15. Consider the following code segment.

```
int a = 1;
int b = 2;
int c = 3;
int d = 4;
double x = a + b * c % d;
```

What is the value of x after executing the code segment?

(A)

1.0

(B)

2.5

(C) ✓

3.0

(D)

7.0

Answer C

Correct. The multiplication b * c is performed first and evaluates to 6. The remainder operation 6 % d is performed next and evaluates to 2. The addition operation a + 2 is performed last and evaluates to 3.

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16. Consider the following code segment.

```
int x = 10;
int y = 20;
/* missing code */
System.out.print(top / bottom);
```

Which of the following replacements for `/* missing code */` will cause an `ArithmeticException` to occur?

I.
```
int top = x - y;
int bottom = y - x;
```

II.
```
int top = 2 * x;
int bottom = y - top;
```

III.
```
int top = x + y;
int bottom = 2 * top;
```

(A) I only

(B) II only ✓

(C) III only

(D) II and III

Answer B

Correct. The correct answer is II only. In statement I, the value of `top` is `-10`, the value of `bottom` is `10`, the integer division `-10 / 10` evaluates to `-1`, and the code segment does not cause any errors or exceptions. Statement II throws an `ArithmeticException` error because the value of `top` is `20` and the value of bottom is `0`, which leads to integer division by `0`. In statement III, the value of `top` is `30`, the value of bottom is `60`, the integer division `30 / 60` evaluates to `0`, and the code segment does not cause any errors or exceptions.

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17. Consider the following statement.

```
System.out.print("AP \n Computer \n Science \n A \n rocks");
```

Which of the following best describes the behavior of this statement?

(A) It prints five lines of text, each separated by a line break indicated by the escape sequence \n. ✓

(B) It prints four lines of text because the escape sequence \n appears four times in the string literal.

(C) It prints one line of text because the printed value is a single string literal.

(D) It prints nothing due to a syntax error.

Answer A

Correct. Each newline escape sequence (\n) inside the string literal causes a line to print.

18. Consider the following code segment.

```
int a = 5;
int b = 2;
double c = 3.0;
System.out.println(5 + a / b * c - 1);
```

What is printed when the code segment is executed?

(A) `9.0`

(B) `10.0` ✓

(C) `11.5`

(D) `14.0`

Answer B

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19. Which of the following arithmetic expressions evaluates to `1` ?

I. `2 / 5 % 3`
II. `2 / (5 % 3)`
III. `2 / 5 + 1`

(A) I only
(B) II only
(C) I and II only
(D) II and III only ✓

Answer D

Correct. Expression I evaluates to `0` since `2 / 5` evaluates to `0` and `0 % 3` also evaluates to `0`. Expression II evaluates to `1` since `5 % 3` evaluates to `2` and `2 / 2` evaluates to `1`. Expression III evaluates to `1` since `2 / 5` evaluates to `0` and `0 + 1` evaluates to `1`.

20. Consider the following code segment.

```
System.out.print("Hello System.out.println");
System.out.print("!!!");
```

What is printed as a result of executing the code segment?

(A) `Hello!!!`
(B) `Hello System.out.println!!!` ✓
(C)
```
Hello
!!!
```
(D)
```
Hello System.out.println
!!!
```

Answer B

Correct. The first print statement prints the entire string literal contained in quotation marks: `"Hello System.out.println"`. The cursor does not move to a new line. The second print statement prints the string literal `"!!!"`.

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21. Consider the following code segment.

```
double firstDouble = 2.5;
int firstInt = 30;
int secondInt = 5;
double secondDouble = firstInt - secondInt / firstDouble + 2.5;
```

What value will be assigned to `secondDouble` when the code segment is executed?

(A) `5.0`

(B) `12.5`

(C) `25.5`

(D) `30.5` ✓

Answer D

Correct. Division has higher precedence than addition and subtraction, which have the same precedence. Operators with the same precedence are evaluated in left-to-right order. The statement that assigns a value to `secondDouble` evaluates `secondInt / firstDouble` first, resulting in `2.0`. The expression `firstInt - 2.0` is evaluated next, resulting in `28.0`. Finally, the expression `28.0 + 2.5` is evaluated resulting in `30.5`.

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22. Consider the following code segment, which is intended to print the digits of the two-digit `int` number `num` in reverse order. For example, if `num` has the value `75`, the code segment should print `57`. Assume that `num` has been properly declared and initialized.

```
/* missing code */
System.out.print(onesDigit);
System.out.print(tensDigit);
```

Which of the following can be used to replace `/* missing code */` so that the code segment works as intended?

(A) ✓

```
int onesDigit = num % 10;

int tensDigit = num / 10;
```

(B)

```
int onesDigit = num / 10;

int tensDigit = num % 10;
```

(C)

```
int onesDigit = 10 / num;

int tensDigit = 10 % num;
```

(D)

```
int onesDigit = num % 100;

int tensDigit = num / 100;
```

Answer A

Correct. The expression `num % 10` extracts the rightmost digit of `num` by evaluating to the remainder when `num` is divided by `10`. The expression `num / 10` extracts the leftmost digit by evaluating to the result of the integer division of `num` by `10`.

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23. Consider the following code segment.

```
System.out.print("AP");
System.out.println();
System.out.println("CS");
System.out.print("A");
```

What is printed as a result of executing the code segment?

(A)
```
APCS
A
```

(B)
```
AP
CSA
```

(C) ✓
```
AP
CS
A
```

(D)
```
AP

CS
A
```

Answer C

Correct. The first statement prints `"AP"`. The second statement prints no characters but moves the cursor to a new line. The third statement prints `"CS"` and moves the cursor to a new line. The last statement prints `"A"`.

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24. Consider the following code segment.

```
System.out.print(I do not fear computers. );   // Line 1
System.out.println(I fear the lack of them.);  // Line 2
System.out.println(--Isaac Asimov);            // Line 3
```

The code segment is intended to produce the following output but may not work as intended.

```
I do not fear computers. I fear the lack of them.
--Isaac Asimov
```

Which change, if any, can be made so that the code segment produces the intended output?

(A) In line 1, print should be changed to println.

(B) The statement System.out.println() should be inserted between lines 2 and 3.

(C) In lines 1, 2, and 3, the text that appears in parentheses should be enclosed in quotation marks. ✓

(D) No change is needed; the code segment works correctly as is.

Answer C

Correct. For the text that appears in parentheses to be interpreted as string literals, it must be enclosed in quotation marks.

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25. Consider the following code segment.

```
System.out.print(*);      // Line 1
System.out.print("*");    // Line 2
System.out.println();     // Line 3
System.out.println("*"); // Line 4
```

The code segment is intended to produce the following output, but may not work as intended.

```
**
*
```

Which line of code, if any, causes an error?

(A) Line 1 ✓

(B) Line 2

(C) Line 3

(D) The code segment works as intended.

Answer A

Correct. For line 1 to work as intended, the asterisk must be enclosed in quotation marks so it is interpreted as a string literal.

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26. Each of the following code segments is intended to print the word `Hello`. Which of the following code segments works as intended?

I.
```
System.out.print("Hello");
```
II.
```
System.out.print(Hello);
```
III.
```
System.out.print(He);
System.out.print(llo);
```

(A) I only ✓

(B) III only

(C) I and II

(D) II and III

Answer A

Correct. Code segment I correctly prints the string literal `"Hello"`. Code segment II does not work as intended because `Hello` is not enclosed in quotation marks. Code segment III does not work as intended because `He` and `llo` are not enclosed in quotation marks.

FINAL Unit1 MCQ - Topic order - Deduplicated

27. Which of the following code segments will print the word Hello?

(A) ✓

```
System.out.print("Hello");
```

(B)

```
System.out.print(Hello);
```

(C)

```
"System.out.print(Hello)";
```

(D)

```
System.out.print(He);

System.out.print(llo);
```

Answer A

Correct. The code segment correctly prints the string literal "Hello".

FINAL Unit1 MCQ - Topic order - Deduplicated

28. Consider the following code segment.

```
System.out.print("*");
System.out.println("**");
System.out.println("***");
System.out.print("****");
```

What is printed as a result of executing the code segment?

(A)
```
*
**
***
****
```

(B)
```
*
**
*******
```

(C) ✓
```
***
***
****
```

(D)
```
***
*******
```

Answer C

Correct. The arguments of the first and second statements appear on the first line of output, since the `print` statement in line 1 of the code segment does not move the cursor to the next line before the `println` statement in line 2 of the code segment is executed. This produces a line of output containing `***` and then moves the cursor to the next line. The `println` statement in line 3 of the code segment produces another line of output containing `***` and then moves the cursor to the next line. The final `print` statement produces a third line of output that contains `****`.

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29. Consider the following code segment.

```
System.out.print("One");   // Line 1
System.out.print("Two");   // Line 2
System.out.print("Three"); // Line 3
System.out.print("Four");  // Line 4
```

The code segment is intended to produce the following output, but does not work as intended.

```
OneTwo
ThreeFour
```

Which of the following changes can be made so that the code segment produces the intended output?

(A) Changing `print` to `println` in line 1 only

(B) Changing `print` to `println` in line 2 only ✓

(C) Changing `print` to `println` in line 3 only

(D) Changing `print` to `println` in lines 1, 2, 3, and 4

Answer B

Correct. As is, the code segment prints all four strings on the same line. Changing `print` to `println` in line 2 will move the cursor to the next line after `"Two"` is printed.

30. Which of the following statements stores the value `3` in `x` ?

(A) `int x = 7 / 3;`

(B) `int x = 7 / 4;`

(C) `int x = 5 % 8;`

(D) `int x = 8 % 5;` ✓

Answer D

Correct. The expression `8 % 5` evaluates to the remainder left when `8` is divided by `5`, or `3`.

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31. Consider the following code segment.

```
int a = 3 + 2 * 3;
int b = 4 + 3 / 2;
int c = 7 % 4 + 3;
double x = a + b + c;
```

What is the value of x after the code segment is executed?

(A)

14.0

(B)

18.0

(C) ✓

20.0

(D)

20.5

Answer C

Correct. In the first assignment statement, the multiplication operation is evaluated before the addition operation and a is assigned the value 9. In the second assignment statement, the integer division is evaluated first and produces a result of 1, which is added to 4 so that the variable b is assigned the value 5. In the third assignment statement, the remainder operation is evaluated before the addition operation and c is assigned the value 6. The variable x is assigned the value 20.0.

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32. In the following code segment, `x` and `y` are properly declared and initialized `double` variables.

```
y = x;
double z = y + x;
System.out.println(z);
```

Which of the following best describes what is printed as a result of executing the code segment?

(A) Two times the initial value of `x` ✓

(B) Two times the initial value of `y`

(C) The sum of the initial values of `x` and `y`

(D) The value of `x` followed by the value of `y`

Answer A

Correct. The code segment assigns the value of `x` to `y`. Therefore, the sum `y + x` is equivalent to `x + x`, or two times the initial value of `x`.

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33. Consider the following code segment.

```
int w = 50;
int x = 20;
int y = x;
x = w;
w = x + y;
System.out.println(w + " " + x);
```

What is printed as a result of executing the code segment?

(A)

40 20

(B)

70 20

(C) ✓

70 50

(D)

100 50

Answer C

Correct. In this code segment, w is initially assigned 50 and x is initially assigned 20. Next, y is assigned the value of x, which is 20. Next, x is assigned the value of w, which is 50. Next, w is assigned the value of x + y, which is 50 + 20, or 70. Last, the code segment prints the values of w and x.

FINAL Unit1 MCQ - Topic order - Deduplicated

34. Consider the following code segment.

```
int first = 5 + 10 * 2;
int second = first + first % 2;
```

What is the value of second after this code segment is executed?

(A)

37

(B)

30

(C) ✓

26

(D)

0

Answer C

Correct. In the first statement, multiplication takes precedence over addition. The first statement multiplies 10 times 2, and then adds 5. Therefore, 25 is assigned to first. In the second statement, remainder (%) takes precedence over addition. The second statement calculates the remainder when 25 is divided by 2 (which is 1) and then adds 25. Therefore, 26 is assigned to second.

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35. Consider the following code segment.

```
int x = 20;
int y = 10;
int temp = x;
x = y;
y = temp * x;
System.out.println(x + " " + y);
```

What is printed as a result of executing this code segment?

(A)

10 100

(B) ✓

10 200

(C)

20 100

(D)

20 200

Answer B

Correct. This code segment initially assigns 20 to x and 10 to y. Next, the value of x, which is 20, is assigned to temp. Next, the value of y, which is 10, is assigned to x. Next, the value of temp * x, which is equal to 10 * 20, or 200, is assigned to y. The final values of x and y are 10 and 200.

FINAL Unit1 MCQ - Topic order - Deduplicated

36. Which of the following expressions evaluates to 3.5?

(A) ✓

```
(double) 2 / 4 + 3
```

(B)

```
(double) 2 / (4 + 3)
```

(C)

```
(double) (2 / 4) + 3
```

(D)

```
(double) (2 / 4 + 3)
```

Answer A

Correct. In this expression, the cast applies to the value 2, so double division is performed and the expression evaluates to 0.5 + 3, or 3.5.

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37. Consider the following code segment.

```
double a = 7;
int b = (int) (a / 2);
double c = (double) b / 2;
System.out.print(b);
System.out.print(" ");
System.out.print(c);
```

What is printed as a result of executing the code segment?

(A)

3 1.0

(B) ✓

3 1.5

(C)

3.5 1.5

(D)

3.5 1.75

Answer B

Correct. The value of a / 2 is 3.5. When cast to an int, the digits to the right of the decimal point are truncated, so b contains the value 3. The value c is calculated by dividing 3.0 by 2, resulting in 1.5.

FINAL Unit1 MCQ - Topic order - Deduplicated

38. Consider the following code segment.

```
double q = 15.0;
int r = 2;

double x = (int) (q / r);
double y = q / r;

System.out.println(x + " " + y);
```

What is printed as a result of executing this code segment?

(A)

7.0 7.0

(B) ✓

7.0 7.5

(C)

7.5 7.0

(D)

7.5 7.5

Answer B

Correct. The value of x is 7.0. The expression 15.0 / 2 evaluates to 7.5, which is cast to the int value 7 and then implicitly cast to the double value 7.0. The value of y is 7.5 because the expression 15.0 / 2 evaluates to 7.5 and no casting is applied to this result.

FINAL Unit1 MCQ - Topic order - Deduplicated

39. Consider the following code segment.

```
double a = 1.1;
double b = 1.2;

if ((a + b) * (a - b) != (a * a) - (b * b))
{
  System.out.println("Mathematical error!");
}
```

Which of the following best describes why the phrase `"Mathematical error!"` would be printed?

(Remember that mathematically $(a + b) * (a - b) = a^2 - b^2$.)

(A) Precedence rules make the `if` condition true.

(B) Associativity rules make the `if` condition true.

(C) Roundoff error makes the `if` condition true. ✓

(D) A compiler bug or hardware error has occurred.

FINAL Unit1 MCQ - Topic order - Deduplicated

40. Consider the following code segment.

```
int a = 5;
int b = 4;
int c = 2;
a *= 3;
b += a;
b /= c;
System.out.print(b);
```

What is printed as a result of executing the code segment?

(A)

4

(B) ✓

9

(C)

9.5

(D)

19

Answer B

Correct. The variable a is initialized to 5 and then multiplied by 3, resulting in 15. The variable b is initialized to 4 and then incremented by the value of a, resulting in 19. The variable b is then divided by c using integer division. Since c was initialized to 2, the result of the integer division is 9.

FINAL Unit1 MCQ - Topic order - Deduplicated

41. The following code segment appears in a method. In the code segment, y is an int variable.

```
int x = 5 / y;
System.out.println(x);
```

Which of the following preconditions for the method is most appropriate to avoid an ArithmeticException?

(A)

```
/** Precondition: y is equal to 0. */
```

(B)

```
/** Precondition: y is less than 5. */
```

(C) ✓

```
/** Precondition: y is not equal to 0. */
```

(D)

```
/** Precondition: y is not negative. */
```

Answer C

Correct. An ArithmeticException will occur when an integer is divided by 0. Specifying that y is not equal to 0 will prevent this error from occurring.

FINAL Unit1 MCQ - Topic order - Deduplicated

42. The following code segment appears in a method. In the code segment, `x` and `y` are properly declared and initialized `int` variables.

```
System.out.println(x * x * y);
```

Which of the following preconditions for the method is most appropriate to ensure that the value printed by the code segment is always greater than zero?

FINAL Unit1 MCQ - Topic order - Deduplicated

(A)

```
/**
 * Preconditions: x is greater than zero.
 *                y is not equal to zero.
 */
```

(B) ✓

```
/**
 * Preconditions: x is not equal to zero.
 *                y is greater than zero.
 */
```

(C)

```
/**
 * Preconditions: x is even and is not equal to zero.
 *                y is not equal to zero.
 */
```

(D)

```
/**
 * Preconditions: x is not equal to zero.
 *                y is even and is not equal to zero.
 */
```

Answer B

Correct. Whether `x` is positive or negative, `x * x` is positive. Therefore, for the printed value to be positive, `y` must be greater than zero. When `x` is zero, the expression `x * x * y` will evaluate to `0`, which is not

positive. Therefore `x` must not be equal to `0`.

43. In the following code segment, `x` and `y` are properly declared and initialized `int` variables.

```
int z = x;
x = y;
y = z;
```

Which of the following best describes the behavior of the code segment?

(A) It assigns the original value of `x` to both `x` and `y`.

(B) It assigns the original value of `y` to both `x` and `y`.

(C) It leaves the values of `x` and `y` unchanged.

(D) It swaps the values of `x` and `y`. ✓

Answer D

Correct. The first line of code assigns the original value of `x` to the temporary variable `z`. The second line assigns the original value of `y` to `x`. The third line of code assigns the value of `z`, which holds the original value of `x`, to `y`.

FINAL Unit1 MCQ - Topic order - Deduplicated

44. The following code segment contains an error.

```
double pi = 3.14;    // line 1
pi = 3.14159;        // line 2
double other = null; // line 3
double example;      // line 4
example = pi;        // line 5
```

Which of the following best describes the error in this code segment?

(A) In line 2, pi is assigned a value after it was already assigned a value in line 1.

(B) In line 3, null cannot be assigned to a non-reference type. ✓

(C) In line 4, example should be assigned a value.

(D) In line 5, a variable cannot be assigned to another variable.

Answer B

Correct. The keyword null can only be assigned to a reference type, not to a primitive type like double.

FINAL Unit1 MCQ - Topic order - Deduplicated

45. The following code segment is intended to interchange the values of the int variables x and y. Assume that x and y have been properly declared and initialized.

```
int temp = x;
/* missing code */
```

Which of the following can be used to replace /* missing code */ so that the code segment works as intended?

(A) ✓

```
x = y;

y = temp;
```

(B)

```
y = x;

x = temp;
```

(C)

```
y = x;

temp = y;
```

(D)

```
y = x;

temp = x;
```

Answer A

Correct. Since the original value of x has been stored in temp, the variable x can be assigned the value of y and then y can be assigned the original value of x, as stored in temp.

FINAL Unit1 MCQ - Topic order - Deduplicated

46. Consider the following code segment.

```
boolean a = true;
boolean b = false;
boolean temp = a;
a = b;
b = temp;
System.out.println(a);
System.out.println(b);
```

What is printed as a result of executing this code segment?

(A)

true

true

(B)

true

false

(C) ✓

false

true

(D)

false

false

Answer C

Correct. This code segment swaps the values of a and b. It initially assigns true to a and false to b. Next, the value of a, which is true, is assigned to temp. Next, the value of b, which is false, is assigned to a. Next, the value of temp, which is true, is assigned to b. The final values of a and b are false and true.

FINAL Unit1 MCQ - Topic order - Deduplicated

47. The volume of a cylinder is equal to the height times the area of the circular base. The area of the circular base is equal to π (pi) times the square of the radius.

The code segment below is intended to compute and print the volume of a cylinder with radius `r` and height `h`. Assume that the `double` variables `r`, `h`, and `pi` have been properly declared and initialized.

```
/* missing code */
System.out.print(volume);
```

Which of the following can be used to replace `/* missing code */` so that the code segment works as intended?

I.
```
double baseArea = pi * r * r;
double volume = baseArea * h;
```

II.
```
double volume = pi * r * r;
volume = volume * h;
```

III.
```
double volume = pi * r * r * h;
```

(A) I only

(B) I and III only

(C) II and III only

(D) I, II, and III ✓

Answer D

Correct. Code segment I correctly calculates `baseArea` as `pi * r * r`, then correctly calculates `volume` by multiplying `baseArea` by `h`. Code segment II calculates the base area of the cylinder and temporarily stores it in `volume`. This value is multiplied by `h`, resulting in the correct volume. Code segment III correctly calculates the volume by multiplying `pi * r * r * h` in a single expression.

FINAL Unit1 MCQ - Topic order - Deduplicated

48. The volume of a cylinder is equal to the height times the area of the circular base. The area of the circular base is equal to π (pi) times the square of the radius.

The following code segment is intended to compute and print the volume of a cylinder with radius r and height h. Assume that the double variables r, h, and pi have been properly declared and initialized.

```
/* missing code */
System.out.print(volume);
```

Which of the following **cannot** be used to replace /* missing code */ so that the code segment works as intended?

(A) ✓

```
double baseArea;

double volume;

pi * r * r = baseArea;

baseArea * h = volume;
```

(B)

```
double baseArea = pi * r * r;

double volume = baseArea * h;
```

(C)

```
double volume = pi * r * r ;

volume = volume * h;
```

(D)

```
double volume = pi * r * r * h;
```

Answer A

Correct. The code segment uses incorrect assignment statements. The expressions pi * r * r and baseArea * h should be on the right side of the = rather than the left side.

FINAL Unit1 MCQ - Topic order - Deduplicated

49. In the following code segment, `x` and `y` are properly declared and initialized `int` variables whose values are positive. The code segment is intended to print the average of `x` and `y`. For example, if `x` is `10` and `y` is `5`, the code segment is intended to print `7.5`.

```
int sum = x + y;
double average = (double) (sum / 2);
System.out.println(average);
```

Which of the following best describes the error, if any, in the code segment?

(A) There is no error; the code segment works as intended.

(B) In the expression `(double) (sum / 2)`, the cast to `double` is applied too late, so the average will be less than the expected result for even values of `sum`.

(C) In the expression `(double) (sum / 2)`, the cast to `double` is applied too late, so the average will be less than the expected result for odd values of `sum`. ✓

(D) In the expression `(double) (sum / 2)`, the cast to `double` is applied too late, so the average will be greater than the expected result for odd values of `sum`.

Answer C

Correct. Since `sum` is declared as an `int` type and `2`, by default, is also an `int` type, the division will result in an `int` type causing a loss of precision whenever an odd number is divided by an even number. Casting the expression `sum / 2` to `double` after the division has occurred is too late. It is not possible to recover the loss of precision. So for odd values of `sum`, the value of `average` will be less than the expected average.

FINAL Unit1 MCQ - Topic order - Deduplicated

50. Assume that x is a double variable with a positive value. Which of the following code segments can be used to round x to the nearest integer and store the rounded value in the variable result?

(A)

```
int result = (int) x;
```

(B)

```
int result = (int) x + 0.5;
```

(C) ✓

```
int result = (int)(x + 0.5);
```

(D)

```
int result = (int) x + (int) 0.5;
```

Answer C

Correct. This code segment adds 0.5 to x, then casts the result to an int, which removes the digits to the right of the decimal point. For example, if x is 4.2, the code segment adds 4.2 + 0.5, then casts the result 4.7 to the int 4. As another example, if x is 7.6, the code segment adds 7.6 + 0.5, then casts the result 8.1 to the int 8. In both of these examples, the original value of x is rounded to the nearest integer.

FINAL Unit1 MCQ - Topic order - Deduplicated

51. Consider the following code segment.

```
double num = 9 / 4;
System.out.print(num);
System.out.print(" ");
System.out.print((int) num);
```

What is printed as a result of executing the code segment?

(A) ✓

2.0 2

(B)

2.0 2.0

(C)

2.25 2

(D)

2.25 2.0

Answer A

Correct. The value of `num` is `2.0` since `9` and `4` are both `int` values. This results in integer division. The first print statement will display `2.0`. The second print statement will display a space. In the third print statement, `num` is cast to an `int` value, so the statement will display `2`.

FINAL Unit1 MCQ - Topic order - Deduplicated

52. Consider the following code segment.

```
double x = (int) (5.5 - 2.5);
double y = (int) 5.5 - 2.5;
System.out.println(x - y);
```

What is printed as a result of executing the code segment?

(A) `-0.5`

(B) `0.0`

(C) `0.5` ✓

(D) `1.0`

Answer C

Correct. The code segment prints the `double` value `0.5`. The value of the `double` variable `x` is `3.0` because the value of `5.5 - 2.5` is `3.0`, the cast from the `double` value `3.0` to an `int` is `3`, and the implicit cast from the `int` value `3` to a `double` is `3.0`. The value of the `double` variable `y` is `2.5` because the cast from the `double` value `5.5` to an `int` is `5` and the value of `5 - 2.5` is the `double` value `2.5`. The code segment prints `0.5` because the value of `x - y` is `3.0 - 0.5`, which evaluates to `0.5`.

FINAL Unit1 MCQ - Topic order - Deduplicated

53. Consider the following code segment.

```
int w = 1;
int x = w / 2;
double y = 3;
int z = (int) (x + y);
```

Which of the following best describes the results of compiling the code segment?

(A) The code segment compiles without error. ✓

(B) The code segment does not compile, because the `int` variable `x` cannot be assigned the result of the operation `w / 2`.

(C) The code segment does not compile, because the integer value `3` cannot be assigned to the `double` variable `y`.

(D) The code segment does not compile, because the operands of the addition operator cannot be of different types `int` and `double`.

Answer A

Correct. The first statement assigns an integer value to an `int` variable. The second statement performs integer division and assigns the `int` result `0` to an `int` variable. In the third statement, the integer value `3` is cast to a `double` and assigned to a `double` variable. In the final statement, the sum of `x` and `y` is a `double`, since at least one of the operands is a `double`. The `double` result is cast to an `int` and stored in an `int` variable.

FINAL Unit1 MCQ - Topic order - Deduplicated

54. Consider the following code segment.

```
double x = 4.5;
int y = (int) x * 2;
System.out.print(y);
```

What is printed as a result of executing the code segment?

(A) `8` ✓

(B) `8.0`

(C) `9`

(D) `9.0`

Answer A

Correct. The code segment prints the `int` value `8`. In the expression `(int) x * 2`, the value of `double` variable `x` is first cast to an `int`, resulting in `4`. This result is then multiplied by `2`, resulting in `8`.

FINAL Unit1 MCQ - Topic order - Deduplicated

55. Consider the following code segment.

```
double w = 2.0;
double x = 5.0;
double y = (int) w / x;
double z = (int) (w / x);

System.out.println(y + ", " + z);
```

What is printed as a result of executing this code segment?

(A)

0.0, 0.0

(B)

0.0, 0.4

(C) ✓

0.4, 0.0

(D)

0.4, 0.4

Answer C

Correct. The expression (int) w / x first casts w to the int 2 and then divides the result by the double 5.0. When an int is divided by a double, the int is automatically widened to a double, so the expression is calculated as 2.0 / 5.0, which evaluates to 0.4. The expression (int) (w / x) first divides w by x, which evaluates to 0.4. This result is then cast to an int, which evaluates to 0. This value is then automatically widened to 0.0 when it is assigned to the double variable z.

FINAL Unit1 MCQ - Topic order - Deduplicated

56. A store sells rope only in whole-foot increments. Given three lengths of rope, in feet, the following code segment is intended to display the minimum length of rope, in feet, that must be purchased so that the rope is long enough to be cut into the three lengths.

For example, for lengths of 2.8 feet, 3 feet, and 5 feet, the minimum length of rope that must be purchased is 11 feet. For these values, the code segment should display 11. As another example, for lengths of 1.1 feet, 3.2 feet, and 2 feet, the minimum length of rope that must be purchased is 7 feet. For these values, the code segment should display 7.

In the following code segment, len1, len2, and len3 are properly declared and initialized double variables representing the three lengths of rope.

```
double total = len1 + len2 + len3;
int minLength = (int) (total + 0.5);
System.out.print(minLength);
```

Which of the following best describes the behavior of the code segment?

(A) The code segment works as intended for all nonnegative values of len1, len2, and len3.

(B) The code segment works as intended but only when the sum of the three lengths is an integer or the decimal part of the sum of the three lengths is greater than or equal to 0.5. ✓

(C) The code segment does not work as intended for some values of len1, len2, and len3 because these values must be cast to int before they are added together.

(D) The code segment does not work as intended for some values of len1, len2, and len3 because minLength should be declared as a double and then cast to an int in the print statement.

Answer B

Correct. If the sum of the three lengths is an integer, then the exact amount of rope that is needed is returned. If the decimal part of the sum of the three lengths is greater than or equal to 0.5, then the code segment will correctly round the double value up to the nearest integer value. If the decimal part of the sum of the three lengths is less than 0.5, then the code segment will correctly round the double value down to the nearest integer value. However, to print the correct value, the code segment must always round up to the nearest integer to guarantee that the minimum length of rope is long enough to be cut into the three lengths.

FINAL Unit1 MCQ - Topic order - Deduplicated

57. The following code segment is intended to round `val` to the nearest integer and print the result.

```
double val = -0.7;
int roundedVal = (int) (val + 0.5);
System.out.println(roundedVal);
```

Which of the following best describes the behavior of the code segment?

(A) The code segment works as intended.

(B) The code segment does not work as intended because the expression `(val + 0.5)` should be cast to a `double` instead of an `int`.

(C) The code segment does not work as intended because `val` should be cast to an `int` before `0.5` is added to it.

(D) The code segment does not work as intended because the expression `(int) (val + 0.5)` rounds to the nearest integer only when `val` is positive. ✓

Answer D

Correct. One way to correct the code segment is to test whether the value of val is positive and to set the value of roundedVal to (int) (val + 0.5) when the value of val is positive and to (int) (val - 0.5) otherwise.

FINAL Unit1 MCQ - Topic order - Deduplicated

58. The following code segment appears in a method. In the code segment, num1 and num2 are int variables. The method is intended to print the sum of num1 and num2.

```
int result = num1 + num2;
System.out.println(result);
```

Which of the following preconditions for the method is most appropriate to avoid an overflow error?

(A)

```
/** Precondition: num1 and num2 are both positive. */
```

(B)

```
/** Precondition: num1 is not equal to num2 */
```

(C)

```
/** Preconditions: num1 is between Integer.MIN_VALUE
 *                 and Integer.MAX_VALUE, inclusive.
 *                 num2 is between Integer.MIN_VALUE
 *                 and Integer.MAX_VALUE, inclusive.
 */
```

(D) ✓

```
/** Precondition: (num1 + num2) is between Integer.MIN_VALUE
 *                and Integer.MAX_VALUE, inclusive.
 */
```

Answer D

Correct. An overflow error occurs when an expression evaluates to an expression outside the allowable range, which is less than Integer.MIN_VALUE or greater than Integer.MAX_VALUE. An overflow error can be avoided by specifying that the sum must be between these two values.

FINAL Unit1 MCQ - Topic order - Deduplicated

59. In the code segment below, assume that the `int` variables `a` and `b` have been properly declared and initialized.

```
int c = a;
int d = b;
c += 3;
d--;
double num = c;
num /= d;
```

Which of the following best describes the behavior of the code segment?

(A) The code segment stores the value of `(a + 3) / b` in the variable `num`.

(B) The code segment stores the value of `(a + 3) / (b - 1)` in the variable `num`. ✓

(C) The code segment stores the value of `(a + 3) / (b - 2)` in the variable `num`.

(D) The code segment causes a runtime error in the last line of code because `num` is type `double` and `d` is type `int`.

Answer B

Correct. The third line of code increments `c` by `3`, which is equal to `a + 3`. The fourth line of code decrements `d`, which is equal to `b - 1`. The fifth line of code assigns the current value of `c`, `a + 3`, to the variable `num`. The sixth line of code divides `num` by the current value of `d`, `b - 1`, which is equal to `(a + 3) / (b - 1)`.

FINAL Unit1 MCQ - Topic order - Deduplicated

60. Consider the following code segment.

```
int x = 0;              // line 1
x++;                    // line 2
System.out.print(x);    // line 3
x += 1;                 // line 4
System.out.print(x);    // line 5
x *= 2;                 // line 6
System.out.print(x);    // line 7
x *= 4;                 // line 8
System.out.print(x);    // line 9
```

The code segment is intended to produce the following output. The code segment does not work as intended.

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Which of the following changes can be made so that the code segment works as intended?

(A) Interchanging lines 2 and 4

(B) Interchanging lines 6 and 8

(C) Changing line 6 to `x += 2;`

(D) Changing line 8 to `x *= 2;` ✓

Answer D

Correct. In the given code segment, the initial value of `x` is 0. The expressions `x++` adds 1 to `x`, causing 1 to be printed. The expression `x += 1` adds 1 to `x`, causing 2 to be printed. The expression `x *= 2` multiplies `x` by 2, causing 4 to be printed. The expression `x *= 4` multiples `x` by 4, causing 16 to be printed. By changing line 8 to `x *= 2;`, the value of `x` (4 at this point) is multiplied by 2, causing 8 to be printed.

FINAL Unit1 MCQ - Topic order - Deduplicated

61. Consider the following code segment.

```
int x = 0;
x++;
x += 1;
x = x + 1;
x -= -1;
System.out.println(x);
```

What is printed when the code segment is executed?

(A)

1

(B)

2

(C)

3

(D) ✓

4

Answer D

Correct. After x is initialized to 0, each of the next four statements increments x by 1, resulting in 4.

FINAL Unit1 MCQ - Topic order - Deduplicated

62. Consider the following code segment.

```
num += num;
num *= num;
```

Assume that `num` has been previously declared and initialized to contain an integer value. Which of the following best describes the behavior of the code segment?

(A) The value of `num` is the square its original value.

(B) The value of `num` is two times the square of its original value.

(C) The value of `num` is the square of twice its original value. ✓

(D) It cannot be determined without knowing the initial value of `num`.

Answer C

Correct. The statement `num += num;` adds the value of `num` to `num` and stores the result back in `num`, so that `num` is now two times its original value. The statement `num *= num;` then multiplies the value of `num` by `num` and stores the result back in `num`, so that `num` is now the square of twice its original value.

FINAL Unit1 MCQ - Topic order - Deduplicated

63. Consider the following code segment.

```
int x = 5;
x += 6 * 2;
x -= 3 / 2;
```

What is the value of x after the code segment is executed?

(A)

1

(B)

9

(C)

15.5

(D) ✓

16

Answer D

Correct. The variable x is initialized to 5 and then incremented by 12, resulting in 17. It is then decremented by the result of the integer division of 3 by 2, which is 1, resulting in 16.

FINAL Unit1 MCQ - Topic order - Deduplicated

64. Consider the following code segment, where `k` and `count` are properly declared and initialized `int` variables.

```
k++;
k++;
count++;
k--;
count++;
k--;
```

Which of the following best describes the behavior of the code segment?

(A) The code segment leaves both `k` and `count` unchanged.

(B) The code segment increases both `k` and `count` by `2`.

(C) The code segment increases `k` by `4` and `count` by `2`.

(D) The code segment leaves `k` unchanged and increases `count` by `2`. ✓

Answer D

Correct. The variable `k` is unchanged because it is incremented twice and decremented twice, each time by `1`. The variable `count` is increased by `2` because it is incremented twice, each time by `1`.

FINAL Unit1 MCQ - Topic order - Deduplicated

65. Consider the following code segment.

```
int a = 4;
int b = 5;
a++;
b++;
int c = a + b;
a -= 1;
System.out.println(a + c);
```

What is printed as a result of executing the code segment?

(A)

10

(B)

13

(C)

14

(D) ✓

15

Answer D

Correct. The variables `a` and `b` are initialized to `4` and `5`, respectively, then incremented by `1`, resulting in `5` and `6`. The variable `c` is assigned the sum of `a` and `b`, or `11`. The variable `a` is then decremented by `1`, resulting in `4`. Finally, the sum of `a` and `c`, or `15`, is printed.

FINAL Unit1 MCQ - Topic order - Deduplicated

66. Which of the following code segments can be used to print the output 2526?

FINAL Unit1 MCQ - Topic order - Deduplicated

(A) ✓

```
int val = 5;

val *= val;

System.out.print(val);

val++;

System.out.print(val);
```

(B)

```
int val = 5;

val *= val;

System.out.print(val);

val *= 1;

System.out.print(val);
```

(C)

```
int val = 5;

val += val;

System.out.print(val);

val++;

System.out.print(val);
```

FINAL Unit1 MCQ - Topic order - Deduplicated

(D)

```
int val = 5;

val += val;

System.out.print(val);

val *= 1;

System.out.print(val);
```

Answer A

Correct. The statement val *= val; is equivalent to val = val * val;. It multiples val times itself and stores the result, which is 25, in val. The statement val++; is equivalent to val = val + 1;. It adds 1 to val and stores the result, 26, in val.

FINAL Unit1 MCQ - Topic order - Deduplicated

67. The following is an excerpt of a class specification that appears in an API library.

public class Parcel

A Parcel class is used to create objects that represent the status of packages that are delivered to customers.

The Parcel class has status and location variables that hold information about a package's shipping status and location.

The Parcel constructor initializes a Parcel object with its initial status and location.

The updateStatus() and updateLocation() methods are used to update the status and location of a package.

Based on the class specification, which of the following is a true statement about Parcel objects?

(A) status and location are attributes; updateStatus and updateLocation are behaviors. ✓

(B) status and location are behaviors; updateStatus and updateLocation are attributes.

(C) status and updateStatus are attributes; location and updateLocation are behaviors.

(D) status and updateStatus are behaviors; location and updateLocation are attributes.

Answer A

Correct. Attributes hold information related to a class and are stored in variables. Behaviors describe what instances of a class can do and are defined by methods.

FINAL Unit1 MCQ - Topic order - Deduplicated

68. The following are excerpts of two class specifications that appear in an API library.

`PetOwner`

- The `PetOwner` class represents a person who owns a pet that has been a patient at a veterinary office.
- The `PetOwner` class has a `String` variable, `name`, that holds information about the person's name and an `int` variable, `numOfPets`, that represents the number of pets the person owns.
- The `getPersonName()` and `getNumOfPets()` methods are used to get information about the pet owner.

`Cat`

- The `Cat` class represents information about a cat that is a patient at a veterinary office.
- The `Cat` class has a `String` variable, `catName`, that holds information about the cat's name and a `PetOwner` variable, `owner`, that holds information about the cat's owner.
- The `getCatOwner()` and `getCatName()` methods are used to get information about the cat.

Based on the class specifications, which of the following descriptions is most accurate?

(A) A `Cat` object is an attribute of a `PetOwner` object.

(B) A `PetOwner` object is an attribute of a `Cat` object. ✓

(C) The method `getCatName` is a behavior of a `PetOwner` object.

(D) The method `getNumOfPets` is an attribute of a `PetOwner` object.

Answer B

Correct. The variable `owner` is a `PetOwner` object in the `Cat` class. Variables represent attributes of class objects.

FINAL Unit1 MCQ - Topic order - Deduplicated

69. The following is an excerpt of a class specification that appears in an API library.

public class Robot

A Robot specifies a robot that moves around a two-dimensional coordinate grid.

The Robot class has xCoordinate and yCoordinate variables that hold information about the robot's location on the grid.

The Robot constructor initializes a Robot object with the given coordinates.

The move() and rotate() methods are used to modify the robot's location on the grid.

Based on the class specification, which of the following descriptions is accurate?

(A) Robot objects are attributes of the methods move and rotate.

(B) Robot objects are behaviors of the methods move and rotate.

(C) The methods move and rotate represent attributes of Robot objects.

(D) The methods move and rotate represent behaviors of Robot objects. ✓

Answer D

Correct. Behaviors describe what instances of a class can do and are defined by methods.

FINAL Unit1 MCQ - Topic order - Deduplicated

70. The following is an excerpt of a class specification that appears in an API library.

`public class Shirt`

The `Shirt` class is used to create objects that represent shirts.

The `Shirt` class has `color` and `size` variables that hold information about the characteristics of a shirt.

The `Shirt` constructor initializes a `Shirt` object with the given color and size.

The `getColor` and `getSize` methods return information about a shirt.

A `poloShirt` object will be declared as type `Shirt`.

Based on the class specification, which of the following descriptions is accurate?

(A) The `color` and `size` variables are attributes of the `poloShirt` object. ✓

(B) The `color` and `size` variables are behaviors of the `poloShirt` object.

(C) The `poloShirt` object is an attribute of the `getColor` and `getSize` methods.

(D) The `poloShirt` object is a behavior of the `getColor` and `getSize` methods.

Answer A

Correct. Variables are attributes of class objects. The variables hold information related to a class.

FINAL Unit1 MCQ - Topic order - Deduplicated

71. The following is an excerpt of a class specification that appears in an API library.

`public class Song`

A `Song` specifies a piece of music stored in a playlist.

The `Song` class has `title` and `artist` variables that hold information about a song.

The `Song` constructor initializes a `Song` object with the given title and artist.

The `play()` and `stop()` methods are used to control playback of the song.

Based on the class specification, which of the following correctly identifies one attribute and one behavior of the `Song` class?

(A) Attribute: `title`
Behavior: `play()` ✓

(B) Attribute: `play()`
Behavior: `artist`

(C) Attribute: `Song` constructor
Behavior: `artist`

(D) Attribute: `Song` constructor
Behavior: `stop()`

Answer A

Correct. `title` is a variable that stores information (an attribute), and `play()` is a method that defines an action (a behavior).

FINAL Unit1 MCQ - Topic order - Deduplicated

72. The following is an excerpt of a class specification that appears in an API library.

public class Vehicle

The Vehicle class is used to create objects that represent vehicles.

The Vehicle class has make, model, and year variables that hold information about the vehicle's characteristics.

The Vehicle constructor initializes a Vehicle object with the given make, model, and year.

The getMake(), getModel(), and getYear() methods return information about the vehicle.

Based on the class specification, which of the following descriptions is accurate?

(A) Vehicle objects are attributes of the variables make, model, and year.

(B) Vehicle objects are behaviors of the variables make, model, and year.

(C) The variables make, model, and year are attributes of Vehicle objects. ✓

(D) The variables make, model, and year are behaviors of Vehicle objects.

Answer C

Correct. Attributes hold information related to a class and are stored in variables.

FINAL Unit1 MCQ - Topic order - Deduplicated

73. A school administrator has created a Student class. The class contains variables to represent the following.

- An int variable named studentID to represent the student's ID number
- A String variable named studentName to represent the student's name

The school administrator has also created a Parent class. The class contains variables to represent the following.

- A String variable named parentName to represent the parent's name
- A String variable named email to represent the parent's e-mail address

The object penelope will be declared as type Student. The object mrsPatel will be declared as type Parent.

Which of the following descriptions is accurate?

(A) An attribute of the penelope object is email.

(B) An attribute of the penelope object is Parent.

(C) An attribute of the penelope object is Student.

(D) An attribute of the mrsPatel object is email. ✓

Answer D

Correct. The mrsPatel object is an instance of the Parent class. The mrsPatel object has two attributes: parentName and email.

FINAL Unit1 MCQ - Topic order - Deduplicated

74. A teacher has created a Student class. The class contains the following.

- An int variable named grade to represent the student's grade level
- A String variable named name to represent the student's name
- A double variable named average to represent the student's grade point average
- A method named updateAverage that updates the value of average.

The object greg will be declared as type Student.

Which of the following descriptions is accurate?

(A) greg is an instance of the Student class. ✓

(B) greg is an instance of three attributes.

(C) Student is an instance of the greg object.

(D) updateAverage is an instance of the Student class.

Answer A

Correct. The object greg has been declared as type Student, which creates it as an instance of the Student class.

FINAL Unit1 MCQ - Topic order - Deduplicated

75. A method has a precondition that the variable `num` must be a positive integer divisible by `2` or must be greater than `100`. Which of the following values of `num` does **not** satisfy the precondition?

(A)

```
4
```

(B) ✓

```
5
```

(C)

```
104
```

(D)

```
105
```

Answer B

Correct. The value `5` is not divisible by `2` and is not greater than `100`, so it does not meet the precondition.

FINAL Unit1 MCQ - Topic order - Deduplicated

76. The following code segment appears in a method. The method is intended to print the product of 3 and the nonnegative difference between the int variables num1 and num2.

```
System.out.println(3 * (num1 - num2));
```

Which of the following preconditions, if any, is required so that the method works as intended?

(A)

```
/** Preconditions: num1 >= 0, num2 >= 0 */
```

(B) ✓

```
/** Precondition: num1 >= num2 */
```

(C)

```
/** Precondition: num2 >= num1 */
```

(D) No precondition is required.

Answer B

Correct. The expression num1 - num2 represents the nonnegative difference between num1 and num2 only when num1 >= num2.

FINAL Unit1 MCQ - Topic order - Deduplicated

77. The following code segment appears in a method. In the code segment, x and y are double variables.

```
double result = (x - y) / 2.0;
System.out.println(result);
```

Which of the following preconditions for the method is most appropriate to ensure that the value printed by the code segment is always positive?

(A)

```
/** Precondition: x and y are both positive. */
```

(B)

```
/** Precondition: x and y are both even. */
```

(C) ✓

```
/** Precondition: x > y */
```

(D)

```
/** Precondition: x < y */
```

Answer C

Correct. If x > y, then the difference x - y will be positive, and this difference divided by 2.0 will also be positive.

FINAL Unit1 MCQ - Topic order - Deduplicated

78. Consider the following code segment.

```
double result = Math.abs(x);
```

Which of the following statements about the variable x is true?

(A) x may hold any numerical value. ✓

(B) x must be equal to 0.

(C) x must have a value greater than 0.

(D) x must have a value less than 0.

Answer A

Correct. Any numerical int or double value may be used here.

79. Consider the following method.

```
public double myMethod(int a, boolean b)
{  /* implementation not shown */  }
```

Which of the following lines of code, if located in a method in the same class as `myMethod`, will compile without error?

(A) `int result = myMethod(2, false);`

(B) `double result = myMethod(0, false);` ✓

(C) `double result = myMethod(true, 10);`

(D) `double result = myMethod(2.5, true);`

Answer B

Correct. The method `myMethod` requires the first parameter to be an `int` and the second to be a `boolean`, and it returns a value of type `double`. The `int` value `0` is passed to the `int` variable `a`, the boolean value `false` is passed to the boolean variable `b`, and the returned `double` is stored in the `double` variable `result`.

FINAL Unit1 MCQ - Topic order - Deduplicated

80. The `Student` class has been defined to store and manipulate grades for an individual student. The following methods have been defined for the class.

```
/* Returns the sum of all of the student's grades */
public double sumOfGrades()
{  /* implementation not shown */  }
/* Returns the total number of grades the student has received */
public int numberOfGrades()
{  /* implementation not shown */  }
/* Returns the lowest grade the student has received */
public double lowestGrade()
{  /* implementation not shown */  }
```

Which of the following statements, if located in a method in the `Student` class, will determine the average of all of the student's grades except for the lowest grade and store the result in the `double` variable `newAverage`?

(A) `newAverage = sumOfGrades() / (numberOfGrades() - 1);`

(B) `newAverage = sumOfGrades() - lowestGrade() / (numberOfGrades() - 1);`

(C) `newAverage = (sumOfGrades() - lowestGrade()) / numberOfGrades() - 1;`

(D) `newAverage = (sumOfGrades() - lowestGrade()) / (numberOfGrades() - 1);` ✓

Answer D

Correct. To determine the correct result, the lowest grade must be subtracted from the sum of all grades and then the result must be divided by one less than the total number of grades.

FINAL Unit1 MCQ - Topic order - Deduplicated

81. Consider the following method.

```
public void doSomething()
{
 System.out.println("Something has been done");
}
```

Each of the following statements appears in a method in the same class as `doSomething`. Which of the following statements are valid uses of the method `doSomething` ?

I. `doSomething();`
II. `String output = doSomething();`
III. `System.out.println(doSomething());`

(A) I only ✓

(B) I and II only

(C) I and III only

(D) I, II, and III

Answer A

Correct. The method `doSomething()` is defined as having the return type `void`; therefore, it cannot be used as part of an expression as in statements II and III.

FINAL Unit1 MCQ - Topic order - Deduplicated

82. Consider the following class definition.

```
public class ExamScore
{
 private String studentId;
 private double score;
 public ExamScore(String sid, double s)
 {
   studentId = sid;
   score = s;
 }
 public double getScore()
 {
   return score;
 }
 public void bonus(int b)
 {
   score += score * b/100.0;
 }
}
```

Assume that the following code segment appears in a class other than `ExamScore`.

```
ExamScore es = new ExamScore("12345", 80.0);
es.bonus(5);
System.out.println(es.getScore());
```

What is printed as a result of executing the code segment?

(A) `4.0`

(B) `80.0`

(C) `84.0` ✓

(D) `85.0`

Answer C

Correct. The first line of the code segment correctly declares and initializes an `ExamScore` object `es`. The `studentId` and `score` attributes of the `es` object are initialized to `"12345"` and `80.0`, respectively. The second line of the code segment correctly calls the `bonus` method of the `es` object, which increases the value of the `es` object's `score` attribute by 5 percent, which is `4`. The third line of the code segment correctly calls the `getScore` method of the `es` object and uses the `println` method to display the new value of `84.0`.

FINAL Unit1 MCQ - Topic order - Deduplicated

83. Consider the following description of method `fahrenheitToCelsius`.

`public double fahrenheitToCelsius(double f)` – Takes as input a temperature in degrees Fahrenheit and returns the corresponding temperature in degrees Celsius.

Assume that each of the following code segments appears in a method in the same class as `fahrenheitToCelsius`. Which of the following code segments prints the temperature in degrees Celsius that corresponds to 32 degrees Fahrenheit?

(A)
```
double f = 32.0;
f = fahrenheitToCelsius();
System.out.println(f);
```

(B)
```
double f = 32.0;
fahrenheitToCelsius(f);
System.out.println(f);
```

(C)
```
double f = 32.0;
double c = fahrenheitToCelsius();
System.out.println(c);
```

(D) ✓
```
double f = 32.0;
double c = fahrenheitToCelsius(f);
System.out.println(c);
```

Answer D

Correct. The non-void method `fahrenheitToCelsius` takes a `double` value as a parameter and returns a `double` value. The returned value is stored in the `double` variable `c`, which is then printed.

FINAL Unit1 MCQ - Topic order - Deduplicated

84. Consider the following descriptions of two methods that appear in the same class.

Method Signature	Explanation
public void methodA(int arg)	Calls methodB with the value of arg * 10
public void methodB(int arg)	Displays the value of arg + 10

Consider the call methodA(4), which appears in a method in the same class. What, if anything, is printed as a result of the call methodA(4)?

(A)

40

(B) ✓

50

(C)

140

(D) Nothing is printed.

Answer B

Correct. The value 4 is passed to methodA and stored in methodA's parameter arg. This value is multiplied by 10. The result 40 is then passed to methodB and stored in methodB's parameter arg. The value 10 is added to arg, and the result 50 is printed.

FINAL Unit1 MCQ - Topic order - Deduplicated

85. Consider the following description of method `min`.

`public int min(int first, int second)` – Returns the lesser of its two parameters.

Assume that each of the following expressions appears in a method in the same class as the method `min`. Assume also that the `int` variables `p`, `q`, and `r` have been properly declared and initialized. Which of the following expressions evaluates to the minimum value among `p`, `q`, and `r`?

I. `min(min(p, q), r)`
II. `min(p, min(q, r))`
III. `min(min(p, q), p)`

(A) I only

(B) II only

(C) I and II only ✓

(D) I, II, and III

Answer C

Correct. One way to determine the minimum of three values is to compare two values and then compare the lesser of the two with the third value. The order of comparisons does not matter, so expressions I and II will both return the correct result. Expression III will not always work correctly. Because it involves comparisons between only two of the three values (`p` and `q`), expression III will return a wrong result when `r` is the minimum value.

FINAL Unit1 MCQ - Topic order - Deduplicated

86. Consider the following description of method printSomething.

Method Signature	Explanation
public void printSomething(int num, boolean val)	Prints the value of val immediately followed by the value of num - 1.

Consider the following code segment, which appears in the same class as printSomething.

```
printSomething(1, true);
printSomething(2, true);
```

What is printed as a result of executing the code segment?

(A)

0true1true

(B)

1true2true

(C) ✓

true0true1

(D)

true1true2

Answer C

Correct. The first call to printSomething passes 1 to the parameter num and true to the parameter val. The value of num is decreased by 1, then val and num are each printed, resulting in the output true0. The second call to printSomething passes 2 to the parameter num and true to the parameter val. The value of num is decreased by 1, then val and num are each printed, resulting in the output true1.

FINAL Unit1 MCQ - Topic order - Deduplicated

87. Consider the following method signatures.

Method Signature	Explanation
public void printSum(int x, double y)	Prints the sum of `x` and `y`
public void printProduct(double x, int y)	Prints the product of `x` and `y`

Consider the following code segment.

```
int num1 = 5;
double num2 = 10.0;
printSum(num1, num2);
System.out.println();
printProduct(num1, num2);
```

What, if anything, is printed as a result of executing the code segment?

(A)

15

50

(B)

15

50.0

(C)

15.0

50.0

(D) Nothing is printed because the code does not compile. ✓

FINAL Unit1 MCQ - Topic order - Deduplicated

Answer D

Correct. The method `printProduct` requires that the second argument be an `int`, but the code segment passes it `num2`, which is of type `double`. This argument type mismatch will cause a compiler error.

FINAL Unit1 MCQ - Topic order - Deduplicated

88. Consider the following description of method printValues.

Method Signature	Explanation
public void printValues(int numTimes, int val)	Prints the value of val a total of numTimes times, then moves the cursor to a new line.

Consider the following code segment, which appears in the same class as printValues.

```
printValues(2, 3);
printValues(4, 5);
```

What is printed as a result of executing the code segment?

(A)

222

44444

(B)

2222

33333

(C) ✓

33

5555

(D)

44

555

Answer C

FINAL Unit1 MCQ - Topic order - Deduplicated

Correct. The first call to printValues passes 2 to the parameter numTimes and 3 to the parameter val. This call results in the output 33. The second call to printValues passes 4 to the parameter numTimes and 5 to the parameter val. This call results in the output 5555.

FINAL Unit1 MCQ - Topic order - Deduplicated

89. Consider the following description of method showResult.

Method Signature	Explanation
public void showResult(int x, int y)	Prints the value of y - x

Consider the following code segment, which appears in the same class as showResult.

```
showResult(1, 3);
System.out.print(" ");
showResult(4, 10);
```

What is printed as a result of executing the code segment?

(A)

-6 -2

(B)

-2 -6

(C) ✓

2 6

(D)

6 2

Answer C

Correct. The first call to showResult passes 1 to the parameter x and 3 to the parameter y. This call calculates 3 - 1 and generates the output 2. The call to System.out.print outputs a space after the 2. The second call to showResult passes the value 4 to the parameter x and 10 to the parameter y. This call calculates 10 - 4 and generates the output 6.

FINAL Unit1 MCQ - Topic order - Deduplicated

90. Consider the following method signatures and corresponding descriptions. Assume that both methods appear in the same class.

Method Signature	Description
public int methodOne(int i, int j)	Returns the value of `i + j`
public int methodTwo(int m, int n)	Returns the value of `n - m`

Consider the following code segment, which appears in the same class as `methodOne` and `methodTwo`.

```
int x = methodOne(4, 5) * methodTwo(1, 3);
System.out.println(x);
```

What is printed as a result of executing the code segment?

(A)

-18

(B)

-4

(C)

4

(D) ✓

18

Answer D

Correct. The call to `methodOne(4, 5)` returns `9` and the call to `methodTwo(1, 3)` returns `2`. The result of the multiplication of `9` and `2` is `18`.

FINAL Unit1 MCQ - Topic order - Deduplicated

91. Consider the following description of the `MagicNumber` class which includes one constructor and two methods.

`public MagicNumber()` – Constructs a `MagicNumber` object that represents a number

`public void displayNumber()` – Displays the number to the screen

`public void add_2()` – Increases the number by 2

When located in a method in a class other than `MagicNumber`, which of the following code segments will compile without error?

I.
```
MagicNumber.add_2();
MagicNumber.displayNumber();
```
II.
```
MagicNumber n1 = new MagicNumber();
n1.add_2();
n1.displayNumber();
```
III.
```
n2.add_2();
n2.displayNumber();
```

(A) I only

(B) II only ✓

(C) II and III only

(D) None of the code segments will compile.

Answer B

Correct. Code segment I will not compile. To call a nonstatic method outside the `MagicNumber` class, a `MagicNumber` object must be created and the nonstatic method must be called using the dot operator on the object name. Code segment II will compile successfully, as it first creates an object of the class `MagicNumber` and then uses the dot operator to call nonstatic methods that manipulate an instance variable of that object. Code segment III will not compile because the variable `n2` has not been properly declared and initialized.

FINAL Unit1 MCQ - Topic order - Deduplicated

92. Consider the following description of the `Purchase` class which includes one constructor and one method.

`public Purchase(double purchaseAmt, double taxAmt)` – Constructs a `Purchase` object that represents a single purchase with purchase amount `purchaseAmt` and tax `taxAmt`.

`public void totalAmount()` – Displays the sum of the purchase amount and the tax

Assume that a `Purchase` object `p` has been properly declared and initialized. Which of the following code segments will successfully print the total purchase amount associated with `p`?

(A) `Purchase.totalAmount();`

(B) `System.out.print(p);`

(C) `System.out.print(p.totalAmount());`

(D) `p.totalAmount();` ✓

Answer D

Correct. The dot operator is needed to invoke a non-static method of a class on an object of that class. This code segment will correctly print the purchase amount associated with `p`.

FINAL Unit1 MCQ - Topic order - Deduplicated

93. Consider the following Painting class.

```
public class Painting
{
  private int year;
  private String artist;
  private String title;

  public Painting(int y, String a)
  {
    year = y;
    artist = a;
    title = "Untitled";
  }

  // There are no other constructors.
}
```

Which of the following code segments, appearing in a class other than Painting, will correctly create an instance of a Painting object?

(A)

```
Painting p = new Painting("Frida Kahlo", "Self Portrait", 1939);
```

(B)

```
Painting p = new Painting(1939, "Frida Kahlo", "Self Portrait");
```

(C)

```
Painting p = new Painting("Frida Kahlo", 1939);
```

(D) ✓

```
Painting p = new Painting(1939, "Frida Kahlo");
```

Answer D

Correct. This statement uses the keyword new followed by a call to the Painting constructor. Even though the class has three instance variables, its constructor has only two parameters: an int parameter and a String parameter, in that order. This constructor call has arguments of the appropriate type in the appropriate order.

FINAL Unit1 MCQ - Topic order - Deduplicated

94. Consider the following Date class.

```
public class Date
{
   private String month;
   private int day;

   public Date(String m, int d)
   { /* implementation not shown */ }

   // Class contains no other constructors. Other methods are not shown.
 }
```

Which of the following code segments, appearing in a class other than Date, will correctly create an instance of a Date object?

(A)

```
Date birthday = new Date("September", "5th");
```

(B) ✓

```
Date birthday = new Date("September", 5);
```

(C)

```
Date birthday = new Date("September 5");
```

(D)

```
Date birthday = new Date();

birthday.month = "September";

birthday.day = 5;
```

Answer B

Correct. A Date object can be created using the keyword new followed by a call to the Date constructor. The constructor requires a String as its first parameter and an int as its second parameter.

FINAL Unit1 MCQ - Topic order - Deduplicated

95. The following method header appears in the `Quadrilateral` class.

```
/**
 *  Returns the area of a quadrilateral with
 *  length len and width wid
 */
public static double findArea(double len, double wid)
```

Which of the following code segments, when appearing in a class other than `Quadrilateral`, will compile without error?

(A)

```
int area = findArea(11.0, 8.5);
```

(B)

```
double area = findArea(11.0, 8.5);
```

(C)

```
int area = Quadrilateral.findArea(11.0, 8.5);
```

(D) ✓

```
double area = Quadrilateral.findArea(11.0, 8.5);
```

Answer D

Correct. The method header includes the keyword `static`, which means that `findArea` is a class method. Class methods are called appropriately using the class name along with the dot operator. The method's return value is stored in a variable of a compatible type.

FINAL Unit1 MCQ - Topic order - Deduplicated

96. Consider the following information about the calculateBill method.

- The method is a static method in the Store class.
- The method has one int parameter.
- The method has return type double.
- The method can be accessed from another class.

Which of the following code segments, when appearing in a method in a class other than Store, will compile without error?

(A)

```
double tot = calculateBill(5);
```

(B) ✓

```
double tot = Store.calculateBill(5);
```

(C)

```
double tot;

tot.calculateBill(5);
```

(D)

```
double tot = 5.0;

tot.calculateBill();
```

Answer B

Correct. Since the method is static, it is called using the class name Store along with the dot operator. The method call uses a valid int argument (5). The method's return value is assigned to a variable that is the same as the method's return type (double).

FINAL Unit1 MCQ - Topic order - Deduplicated

97. Consider the following information about the getTax method.

- The method is a static method in the PurchaseManager class.
- The method has one double parameter.
- The method returns a double equal to its parameter times 0.08.

The following code segment appears in another method in the PurchaseManager class.

```
double tax = getTax(100.0);
System.out.println(tax);
```

What, if anything, is printed as a result of executing the code segment?

(A) 8.0 ✓

(B) 108.0

(C) Nothing is printed because the method call should be PurchaseManager.getTax(100.0).

(D) Nothing is printed because the return value from getTax cannot be assigned to the variable tax.

Answer A

Correct. The method returns its parameter 100.0 times 0.08, which is 8.0. Since the call to this static method appears in the same class, the class name PurchaseManager is not needed in the call.

FINAL Unit1 MCQ - Topic order - Deduplicated

98. Consider the following information about the generatePasscode method.

- The method appears in the AccountManager class.
- The method has no parameters.
- The method has return type int.
- The method can be accessed from another class.

The following code segment appears in a method in a class other than AccountManager class.

```
int c = AccountManager.generatePasscode();
System.out.println(c);
```

Which of the following must be true for the code segment to compile without error?

(A) The generatePasscode method should be designated as void.

(B) The generatePasscode method should be designated as static. ✓

(C) The AccountManager class should contain another method with the name generatePasscode that has at least one parameter.

(D) The AccountManager class should contain another method with the name generatePasscode that has return type double.

Answer B

Correct. Since the method call uses the class name AccountManager along with the dot operator, it is associated with the class, not instances of the class. Class methods are declared using the keyword static.

FINAL Unit1 MCQ - Topic order - Deduplicated

99. The `toCelsius` method appears in the `TemperatureConverter` class. The method takes a parameter that represents a temperature in degrees Fahrenheit and correctly returns the equivalent temperature in degrees Celsius. The header for the `toCelsius` method is shown.

```
/**
 *  Returns the Celsius equivalent of a
 *  Fahrenheit temperature
 */
public static double toCelsius(double fahrenheit)
```

The following code segment appears in a class other than `TemperatureConverter`. It is intended to print the Celsius equivalent of 212 degrees Fahrenheit.

```
double temp = toCelsius(212.0);  // Line 1
System.out.println(temp);
```

Which of the following changes, if any, can be made so that the code segment works as intended?

(A) No change is needed; the code segment works as intended.

(B) Changing line 1 to `double temp.toCelsius(212.0);`

(C) Changing line 1 to `double temp = TemperatureConverter.toCelsius(212.0);` ✓

(D) Changing line 1 to `double temp = toCelsius.TemperatureConverter(212.0);`

Answer C

Correct. The `toCelsius` method is a class method because it is declared using the keyword `static`. It should be called with the class name `TemperatureConverter` along with the dot operator.

FINAL Unit1 MCQ - Topic order - Deduplicated

100. Consider the following description of the `AnimalPrinter` class which includes two methods.

`public void printDog()` – Displays the word "`dog`" and then moves the cursor to a new line

`public void printCat()` – Displays the word "`cat`" and then moves the cursor to a new line

The method `myMethod` appears in a class other than `AnimalPrinter`. The method is intended to produce the following output.

```
dog
cat
```

Assume that an `AnimalPrinter` object `myPrinter` has been properly declared and initialized inside `myMethod`. Which of the following code segments, if located in `myMethod`, will produce the intended output?

(A)
```
printDog();
printCat();
```

(B)
```
printDog(AnimalPrinter);
 printCat(AnimalPrinter);
```

(C)
```
AnimalPrinter.printDog();
 AnimalPrinter.printCat();
```

(D) ✓
```
myPrinter.printDog();
myPrinter.printCat();
```

Answer D

Correct. To work correctly, the dot operator must be used to invoke a non-static method of a class on an object of that class. This code segment correctly invokes the `myPrinter` object's methods `printDog()` and `printCat()` and will produce the desired output.

FINAL Unit1 MCQ - Topic order - Deduplicated

101. The AnimalPrinter class includes the following two methods.

Method Signature	Explanation
public void printDog()	Prints the word "dog" and then moves the cursor to a new line.
public void printCat()	Prints the word "cat" and then moves the cursor to a new line.

The method myMethod appears in a class other than AnimalPrinter. The method is intended to produce the following output.

```
dog
cat
```

Assume that an AnimalPrinter object myPrinter has been properly declared and initialized inside myMethod. Which of the following code segments, if located in myMethod, will produce the intended output?

FINAL Unit1 MCQ - Topic order - Deduplicated

(A)

```
printDog();

printCat();
```

(B)

```
printDog(AnimalPrinter);

printCat(AnimalPrinter);
```

(C)

```
AnimalPrinter.printDog();

AnimalPrinter.printCat();
```

(D) ✓

```
myPrinter.printDog();

myPrinter.printCat();
```

Answer D

Correct. To work correctly, the dot operator must be used to invoke a non-static method of a class on an object of that class. This code segment correctly invokes the myPrinter object's methods printDog() and printCat() and will produce the desired output.

FINAL Unit1 MCQ - Topic order - Deduplicated

102. Consider the following class definition.

```
public class Thing
{
  public void talk()
  {
    System.out.print("Hello ");
  }

  public void name()
  {
    System.out.print("my friend");
  }

  public void greet()
  {
    talk();
    name();
  }
  /* Constructors not shown */
}
```

Which of the following code segments, if located in a method in a class other than `Thing`, will cause the message `"Hello my friend"` to be printed?

(A)
```
Thing a = new Thing();
Thing.talk();
Thing.name();
```

(B)
```
Thing a = new Thing();
Thing.greet();
```

(C) ✓
```
Thing a = new Thing();
a.greet();
```

(D)
```
Thing a = new Thing();
a.name();
a.talk();
```

Answer C

Correct. The statement `Thing a = new Thing();` creates a new `Thing` object called `a`. The statement `a.greet();` uses the dot operator to call the `greet()` method associated with the object `a`. The `greet()` method in turn calls methods `talk()` and `name()`, which print the desired message.

FINAL Unit1 MCQ - Topic order - Deduplicated

103. Consider the following method, which is intended to return true if the distance between d1 and d2 is no more than tolerance units, and to return false otherwise.

For example, almostEqual(7.9, 7.8, 0.15) should return true because the distance between 7.9 and 7.8 is less than 0.15, and almostEqual(54.124, 54.126, 0.001) should return false because the distance between 54.124 and 54.126 is greater than 0.001.

```
public static boolean almostEqual(double d1, double d2,
                                  double tolerance)
{
  /* missing code */;
}
```

Which of the following can be used to replace /* missing code */ so that the method will work as intended?

(A)

```
return (d1 - d2) <= tolerance
```

(B)

```
return ((d1 + d2) / 2) <= tolerance
```

(C) ✓

```
return Math.abs(d1 - d2) <= tolerance
```

(D)

```
return Math.abs((d1 + d2) / 2) <= tolerance
```

Answer C

Correct. The distance between two values is given by the absolute value of their difference. Once the nonnegative difference is calculated, the method will return true if this difference is less than or equal to tolerance and false otherwise.

FINAL Unit1 MCQ - Topic order - Deduplicated

104. Consider the following statement.

```
int b = 5 + (int) (Math.random() * 5);
```

What are the possible values that can be assigned to b as a result of executing the statement?

(A) 5, 6, 7, 8, or 9 ✓

(B) 5, 6, 7, 8, 9, or 10

(C) 6, 7, 8, or 9

(D) 6, 7, 8, 9, or 10

Answer A

Correct. The random method returns a value between 0.0, inclusive, and 1.0, exclusive. Multiplying that value by 5 and casting to an int produces an integer between 0 and 4, inclusive. The sum of 5 and an integer between 0 and 4, inclusive, is an integer value between 5 and 9, inclusive.

FINAL Unit1 MCQ - Topic order - Deduplicated

105.

Consider the following statement, which is intended to assign the value of the expression $\sqrt{\frac{(x+y)^2}{|a-b|}}$ to the variable `result`. Assume that the `double` variables `x`, `y`, `a`, and `b` have been properly declared and initialized.

double result = /* missing code */ ;

Which of the following can replace `/* missing code */` so that the statement works as intended?

(A)

Math.sqrt((x + y) ^ 2 / Math.abs(a, b))

(B)

Math.sqrt((x + y) ^ 2 / Math.abs(a - b))

(C)

Math.sqrt(Math.pow(x + y, 2) / Math.abs(a, b))

(D) ✓

Math.sqrt(Math.pow(x + y, 2) / Math.abs(a - b))

Answer D

Correct. The `Math.pow()` method take two arguments and returns the first argument raised to the power of the second argument. So `Math.pow(x + y, 2)` returns the value of $(x + y)^2$. The `Math.abs()` method takes a single argument and returns the absolute value of the argument. So `Math.abs(a - b)` returns the value of $|a - b|$. Finally, the `Math.sqrt()` method takes a single argument and returns the square root of the argument. So `Math.sqrt(Math.pow(x + y, 2) / Math.abs(a - b))` returns the correct result.

FINAL Unit1 MCQ - Topic order - Deduplicated

106. Which of the following statements assigns a random integer between 25 and 60, inclusive, to `rn` ?

(A) `int rn = (int) (Math.random() * 25) + 36;`

(B) `int rn = (int) (Math.random() * 25) + 60;`

(C) `int rn = (int) (Math.random() * 36) + 25;` ✓

(D) `int rn = (int) (Math.random() * 60) + 25;`

Answer C

107. Which of the following statements assigns a random integer between `1` and `25` inclusive, to `rand`?

(A)

```
int rand = (int) (Math.random()) * 25 + 1;
```

(B)

```
int rand = (int) (Math.random() * 25);
```

(C) ✓

```
int rand = (int) (Math.random() * 25) + 1;
```

(D)

```
int rand = (int) (Math.random() + 1) * 25;
```

Answer C

Correct. The `Math.random()` method returns a `double` that is greater than or equal to `0` and less than `1`. When that value is multiplied by `25`, it evaluates to a `double` that is greater than or equal to `0` and less than `25`. Casting this value to an `int` results in an `int` value from `0` to `24`, inclusive. Adding `1` to that `int` produces a value in the desired range `1` to `25`, inclusive.

FINAL Unit1 MCQ - Topic order - Deduplicated

108. Consider the following code segment, which is intended to assign to `num` a random integer value between `min` and `max`, inclusive, each with an equal chance of being assigned to `num`. Assume that `min` and `max` are integer variables and that the value of `max` is greater than the value of `min`.

```
double rn = Math.random();
int num = /* missing code */ ;
```

Which of the following can replace `/* missing code */` so that the code segment works as intended?

(A)

```
(int) (rn * max) + min
```

(B)

```
(int) (rn * (max - min)) + min
```

(C)

```
(int) (rn * (max - min)) + 1
```

(D) ✓

```
(int) (rn * (max - min + 1)) + min
```

Answer D

Correct. The expression `max - min + 1` represents the number of possible values in the desired range. Multiplying `rn` by the expression `max - min + 1` and casting that product to an `int` produces a value in the range from `0` to `max - min`. Adding `min` to that `int` value produces a value in the desired range of `min` to `max`, inclusive.

FINAL Unit1 MCQ - Topic order - Deduplicated

109. Consider the following code segment.

```
int x = ((int) (Math.random() * 10)) + 5;
int y = ((int) (Math.random() * 5)) + 10;
System.out.println(x + " " + y);
```

Which of the following could be printed as a result of executing this code segment?

(A)

8 15

(B) ✓

9 12

(C)

10 7

(D)

15 11

Answer B

Correct. The expression (int)(Math.random() * (max - min)) + min can be used to produce a random integer greater than or equal to min and less than max. The variable x is assigned a random integer greater than or equal to 5 and less than 15. The value 9 is within this range. The variable y is assigned a random integer greater than or equal to 10 and less than 15. The value 12 is within this range.

FINAL Unit1 MCQ - Topic order - Deduplicated

110. Consider the following code segment.

```
int num = (int)(Math.random() * 6);
System.out.println(num);
```

Which of the following **cannot** be printed as a result of executing this code segment?

(A)

0

(B)

2

(C)

4

(D) ✓

6

Answer D

Correct. The given expression returns an integer greater than or equal to 0, but less than 6. The value 6 is not within this range.

FINAL Unit1 MCQ - Topic order - Deduplicated

111. Consider the following code segment. Assume that `a` is greater than zero.

```
int a = /* value not shown */;
int b = a + (int) (Math.random() * a);
```

Which of the following best describes the value assigned to `b` when the code segment is executed?

(A) `a`

(B) A random integer between `0` and `a - 1`, inclusive

(C) A random integer between `a` and `2 * a`, inclusive

(D) A random integer between `a` and `2 * a - 1`, inclusive ✓

Answer D

Correct. The `random` method returns a value between `0.0`, inclusive, and `1.0`, exclusive. Multiplying that value by `a` and casting to an `int` produces a result between `0` and `a - 1`, inclusive. The sum of `a` and a value between `0` and `a - 1`, inclusive, is a value between `a` and `2 * a - 1`, inclusive.

FINAL Unit1 MCQ - Topic order - Deduplicated

112. Consider the following statement, which is intended to generate an integer between 1 and 10, inclusive, and assign it to result. This statement does not work as intended.

```
int result = (int) Math.random() * 10 + 1;
```

Which of the following changes can be made so that this statement works as intended?

(A) Add parentheses around the expression 10 + 1, as shown in the following statement.

```
int result = (int) Math.random() * (10 + 1);
```

(B) Add parentheses around the expression Math.random() * 10, as shown in the following statement. ✓

```
int result = (int) (Math.random() * 10) + 1;
```

(C) Change the value of 10 to 9, as shown in the following statement.

```
int result = (int) Math.random() * 9 + 1;
```

(D) Remove the cast of Math.random() to an int value, as shown in the following statement.

```
int result = Math.random() * 10 + 1;
```

Answer B

Correct. Adding parentheses around the multiplication ensures that the cast to an int value occurs after the result of the call to Math.random() has been scaled to an appropriate size.

FINAL Unit1 MCQ - Topic order - Deduplicated

113. Assume that the following variable declarations have been made.

```
double d = Math.random();

double r;
```

Which of the following assigns a value to `r` from the uniform distribution over the range `0.5 ≤ r < 5.5` ?

(A) `r = d + 0.5;`

(B) `r = d + 0.5 * 5.0;`

(C) `r = d * 5.0;`

(D) `r = d * 5.0 + 0.5;` ✓

114. Consider the following statement.

```
int r = (int) (Math.random() * 6) + 10;
```

Which of the following best describes the behavior of the statement?

(A) It generates a random integer between 6 and 15, inclusive, and assigns it to r.

(B) It generates a random integer between 6 and 16, inclusive, and assigns it to r.

(C) It generates a random integer between 10 and 15, inclusive, and assigns it to r. ✓

(D) It generates a random integer between 10 and 16, inclusive, and assigns it to r.

Answer C

Correct. The call Math.random() returns a double greater than or equal to 0.0 and less than 1.0. This value is multiplied by 6 to give a double greater than or equal to 0.0 and less than 6.0. Casting this to an int gives an integer between 0 and 5, inclusive. Adding 10 gives an integer between 10 and 15, inclusive.

FINAL Unit1 MCQ - Topic order - Deduplicated

115. The Dog class has a constructor that takes two parameters. The first parameter is a String representing a dog's name and the second parameter is an int representing the dog's age, in months.

The following code segment appears in a class other than Dog.

```
Dog dog1 = new Dog("Rex", 4);
Dog dog2 = dog1;
dog1 = new Dog("Fido", 60);
```

Which of the following best describes the contents of dog1 and dog2 as a result of executing this code segment?

(A) dog1 and dog2 are Dog variables that each contain the same Dog object, which represents a dog named "Fido" that is 60 months old.

(B) dog1 is a Dog variable that contains a Dog object representing a dog named "Fido" that is 60 months old, and dog2 is a Dog variable that contains a Dog object representing a dog named "Rex" that is 4 months old.

(C) dog1 and dog2 are Dog reference variables that each contain a reference to the same memory address that contains the Dog object representing a dog named "Fido" that is 60 months old.

(D) dog1 is a Dog reference variable, which contains a reference to the memory address for the Dog object representing a dog named "Fido" that is 60 months old. dog2 is a Dog reference variable, which contains a reference to the memory address for the Dog object representing a dog named "Rex" that is 4 months old. ✓

Answer D

Correct. Object reference variables store memory locations where objects are stored. The dog1 variable is assigned a new memory location in the third assignment statement.

FINAL Unit1 MCQ - Topic order - Deduplicated

116. A student has created a `Car` class. The class contains the following attributes.

- A `String` variable named `color` to represent the color of the car
- An `int` variable named `year` to represent the year the car was made
- A `String` variable named `make` to represent the manufacturer of the car
- A `String` variable named `model` to represent the model of the car

Assume that `vehicle` is a variable of type `Car` that has been properly declared and initialized. Which of the following best describes a valid relationship?

(A) An instance of the `vehicle` class is `Car`.

(B) An instance of the `Car` object is `vehicle`.

(C) An attribute of the `year` object is `int`.

(D) An attribute of the `vehicle` object is `color`. ✓

Answer D

Correct. The `vehicle` object is an instance of the `Car` class. The `vehicle` object has four attributes: `color`, `year`, `make`, and `model`.

FINAL Unit1 MCQ - Topic order - Deduplicated

117. A student has created an `OrderedPair` class to represent points on an x y-plane. The class contains the following.

- A `double` variable named `x` to represent the x-coordinate of a point
- A `double` variable named `y` to represent the y-coordinate of a point
- A method named `printXY` that prints the values of `x` and `y`

The object `origin` is declared as type `OrderedPair`.

Which of the following statements is true?

(A) `origin` is an instance of the `OrderedPair` class. ✓

(B) `origin` is an instance of two `double` objects.

(C) `OrderedPair` is an instance of the `origin` object.

(D) `printXY` is an instance of the `OrderedPair` class.

Answer A

Correct. The object `origin` is been declared as type `OrderedPair`, which creates it as an instance of the `OrderedPair` class.

FINAL Unit1 MCQ - Topic order - Deduplicated

118. A student has created a `Song` class. The class contains the following three variables.

- A `String` variable named `artist` to represent the artist name
- A `String` variable named `title` to represent the song title
- A `String` variable named `album` to represent the album title

The object `happyBirthday` will be declared as type `Song`.

Which of the following statements is true?

(A) `artist`, `title`, and `album` are instances of the `Song` class.

(B) `happyBirthday` is an instance of three `String` objects.

(C) `happyBirthday` is an instance of the `Song` class. ✓

(D) `Song` is an instance of the `happyBirthday` object.

Answer C

Correct. The object `happyBirthday` is declared as type `Song`, which creates it as an instance of the `Song` class.

FINAL Unit1 MCQ - Topic order - Deduplicated

119. Consider the following class declaration.

```
public class Sample
{
 private int a;
 private double b;

 public Sample(int x, double y)
 {
   a = x;
   b = y;
 }

 // No other constructors
}
```

The following method appears in a class other than `Sample`.

```
public static void test()
{
 Sample object = new /* missing constructor call */ ;
}
```

Which of the following could be used to replace `/* missing constructor call */` so that the method will compile without error?

(A) `Sample()`

(B) `Sample(int x = 10, double y = 6.2)`

(C) `Sample(int x, double y)`

(D) `Sample(10, 6.2)` ✓

Answer D

Correct. The two parameter values are of type `int` and `double`, respectively, and these types match the signature of the only constructor in the `Sample` class.

FINAL Unit1 MCQ - Topic order - Deduplicated

120. Consider the following class definition.

```
public class Student
{
  private String myName;
  private int myAge;

  public Student()
  { /* implementation not shown */ }

  public Student(String name)
  { /* implementation not shown */ }

  public Student(String name, int age)
  { /* implementation not shown */ }

  // There are no other constructors.
}
```

Which of the following statements will cause a compilation error?

(A)

```
Student a = new Student();
```

(B)

```
Student b = new Student("Juan");
```

(C)

```
Student c = new Student("Juan", 15);
```

(D) ✓

```
Student d = new Student("Juan", "15");
```

Answer D

Correct. This statement attempts to use the two-parameter constructor. However, the order of the arguments is a String followed by a String. There is no two-parameter constructor with two String parameters.

FINAL Unit1 MCQ - Topic order - Deduplicated

121. The `WindTurbine` class has two constructors. The constructor headers are shown.

```
public WindTurbine(int efficiency)

public WindTurbine(int efficiency, String location)
```

Which of the following statements, when placed in a method in a class other than `WindTurbine`, will construct a `WindTurbine` object `wt`?

(A) ✓

```
WindTurbine wt = new WindTurbine(3);
```

(B)

```
WindTurbine wt = new WindTurbine(3.25);
```

(C)

```
WindTurbine wt = WindTurbine(3, "Atlantic Ocean");
```

(D)

```
WindTurbine wt =  WindTurbine(3.25, "Atlantic Ocean");
```

Answer A

Correct. The statement invokes the constructor of the `WindTurbine` class with the keyword `new` and an argument of type `int`.

FINAL Unit1 MCQ - Topic order - Deduplicated

122. The Player class has only one constructor. A partial declaration of the constructor is shown.

```
public Player(boolean isOnline, int numLives)
{ /* implementation not shown */ }
```

Which of the following statements correctly creates a Player object?

(A) ✓

```
Player p = new Player(true, 5);
```

(B)

```
Player p = new Player(5, true);
```

(C)

```
Player p = Player(true, 5);
```

(D)

```
Player p = Player(5, true);
```

Answer A

Correct. This is a correct call to instantiate an object of type Player. The call uses the keyword new and uses the name of the class as the type. The call has the correct number of parameters, the parameter types are correct, and the parameters are in the correct order.

FINAL Unit1 MCQ - Topic order - Deduplicated

123. The Person class has exactly two constructors. Partial declarations of the constructors are shown.

```
public Person(int idNumber, boolean isActive)
{ /* implementation not shown */ }

public Person(int idNumber, int age, boolean isActive)
{ /* implementation not shown */ }
```

Which of the following statements does **not** correctly create an object of type Person?

(A)

```
Person p1 = new Person(1138, true);
```

(B)

```
Person p2 = new Person(3136, 17, false);
```

(C) ✓

```
Person p3 = new Person(3544, false, 18);
```

(D)

```
Person p4 = new Person(0, false);
```

Answer C

Correct. This statement attempts to use the second constructor. The list of arguments given is not in the correct order by type. The second argument should be an int and the third argument should be a boolean.

FINAL Unit1 MCQ - Topic order - Deduplicated

124. Consider the following class definition.

```
public class Bird
{
 private String species;
 private String color;
 private boolean canFly;

 public Bird(String str, String col, boolean cf)
 {
   species = str;
   color = col;
   canFly = cf;
 }
}
```

Which of the following constructors, if added to the `Bird` class, will cause a compilation error?

(A)
```
public Bird()
{
 species = "unknown";
 color = "unknown";
 canFly = false;
}
```

(B)
```
public Bird(String col, String str)
{
 species = str;
 color = col;
 canFly = false;
}
```

(C)
```
public Bird(boolean cf, String str, String col)
{
 species = str;
 color = col;
 canFly = cf;
}
```

(D) ✓
```
public Bird(String col, String str, boolean cf)
{
 species = str;
 color = col;
 canFly = cf;
}
```

Answer D

Correct. This constructor has the same signature as the existing constructor (`String, String, boolean`). A

compiler error will occur.

125. Consider the following Point2D class constructor header.

```
public Point2D(double x, double y)
{ /* implementation not shown */ }
```

Which of the following code segments will correctly create an instance of a Point2D object?

(A)

```
Point2D p = Point2D(3.0, 4.0);
```

(B)

```
new p = Point2D(3.0, 4.0);
```

(C)

```
new Point2D = p(3.0, 4.0);
```

(D) ✓

```
Point2D p = new Point2D(3.0, 4.0);
```

Answer D

Correct. This code segment declares a variable p of type Point2D. To initialize the value of p, an instance of the Point2D object is created by using the keyword new followed by a correct call to the class's constructor.

FINAL Unit1 MCQ - Topic order - Deduplicated

126. The `Sofa` class has only two constructors. Partial declarations of the two constructors are shown.

```
public class Sofa(int numOfSeats, double length, String color)
{ /* implementation not shown */ }

public class Sofa(int numOfSeats, double length)
{ /* implementation not shown */ }
```

Which of the following declarations, appearing in a class other than `Sofa`, will correctly create a `Sofa` object?

(A) ✓

```
Sofa s1 = new Sofa(3, 7.5);
```

(B)

```
Sofa s2 = new Sofa(3.0, 8.0);
```

(C)

```
Sofa s3 = new Sofa("3", "7.5", "brown");
```

(D)

```
Sofa s4 = new Sofa(3.0, 8.0, "blue");
```

Answer A

Correct. This declaration correctly uses the second constructor. The arguments in the declaration match the data types in the constructor signature.

FINAL Unit1 MCQ - Topic order - Deduplicated

127. The Vbox class includes the following two constructors.

Constructor Signature	Explanation
public vBox(int w, int h, int d)	Constructs a Vbox object that represents a box with width w, height h, and depth d.
public vBox(int len)	Constructs a Vbox object that represents a box with width len, height len, and depth len.

Which of the following declarations, appearing in a class other than Vbox, will correctly instantiate a Vbox object?

(A)

```
Vbox b1 = new Vbox(4.0);
```

(B)

```
Vbox b2 = new Vbox(4, 2);
```

(C) ✓

```
Vbox b3 = new Vbox(2, 8, 4);
```

(D)

```
Vbox b4 = new Vbox(4.0, 4.0, 4.0);
```

Answer C

Correct. This declaration creates a Vbox object using the constructor with the signature Vbox(int w, int h, int d), which has three int parameters.

FINAL Unit1 MCQ - Topic order - Deduplicated

128. The `Book` class includes the following method.

Method Signature	Explanation
`public int getNumPages(int ch)`	Returns the number of pages in chapter number `ch`

Assume that `Book` object `book1` and `int` variable `c` have been properly declared and initialized in a class other than `Book`. Which of the following statements will compile without error?

(A)

```
int pages = getNumPages(Book book1, int c);
```

(B)

```
int pages = getNumPages(book1, c);
```

(C)

```
int pages = book1.getNumPages(int c);
```

(D) ✓

```
int pages = book1.getNumPages(c);
```

Answer D

Correct. The statement correctly uses the dot operator to call the `getNumPages` method of the `Book` class on the `book1` object and correctly passes the `c` argument to that method.

FINAL Unit1 MCQ - Topic order - Deduplicated

129. The Dog class contains a non-static method named calculateWalkLength which can be called from a class other than Dog. The method takes an integer parameter for the number of minutes a dog is taken for a walk and returns a double representing the number of miles walked during that time.

The following code segment appears in a class other than Dog. The statement in line 2 causes a compile-time error.

```
Dog fido = new Dog();                    // line 1
double miles = calculateWalkLength(fido, 90); // line 2
```

Which of the following can be used to replace line 2 so that this code segment will compile without error?

(A)

```
double miles = calculateWalkLength(90);
```

(B)

```
double miles = Dog.calculateWalkLength(fido, 90);
```

(C)

```
double miles = Dog.calculateWalkLength(90);
```

(D) ✓

```
double miles = fido.calculateWalkLength(90);
```

Answer D

Correct. The method is called with an object of the Dog class and it accepts one integer parameter.

FINAL Unit1 MCQ - Topic order - Deduplicated

130. The Game class includes the following method.

Method Signature	Explanation
public void addPoints(int pts)	Increases the number of points by `pts`

Assume that the `Game` object `myGame` and the `int` variable `points` have been properly declared and initialized in a class other than `Game`. Which of the following code segments is a valid method call?

(A)

```
myGame.addPoints(int points);
```

(B) ✓

```
myGame.addPoints(points);
```

(C)

```
addPoints(Game myGame, int points);
```

(D)

```
addPoints(myGame, points);
```

Answer B

Correct. The statement correctly uses the dot operator to call the `addPoints` method of the `Game` class on the `myGame` object, and correctly passes the points argument to that method.

FINAL Unit1 MCQ - Topic order - Deduplicated

131. Consider the following information about the Person class.

- The class has an int attribute age that represents a person's age.
- The class has a non-static getAge method that has no parameters and returns the value of age. This method can be called from another class.

The following code segments each appear in a class other than Person. Assume that p is a Person object whose age attribute is equal to 16.

Code Segment 1

```
int val1 = p.getAge();
System.out.println(val1);
```

Code Segment 2

```
int val2 = Person.getAge();
System.out.println(val2);
```

What, if anything, is printed as a result of executing each of the code segments?

(A) Both code segments print 16.

(B) Code segment 1 prints 16, but code segment 2 prints nothing because it will not compile. ✓

(C) Code segment 2 prints 16, but code segment 1 prints nothing because it will not compile.

(D) Neither code segment will print anything because they will not compile.

Answer B

Correct. Method getAge is non-static, which means it is called on objects of the Person class along with the dot operator. It cannot be called on the Person class itself.

FINAL Unit1 MCQ - Topic order - Deduplicated

132. Consider the following information about the `Triangle` class.

- The `Triangle` class contains three attributes, `side1`, `side2`, and `side3`, to represent the lengths, in centimeters, of each side of a triangle.
- The class contains one constructor that has three `double` parameters that are used to initialize the attributes of a triangle.
- The class contains an instance method, `perimeter`, which takes no parameters. The method prints the perimeter of a triangle followed by a space (`" "`).

The following code segment appears in a class other than `Triangle`.

```
Triangle isosceles = null;
Triangle scalene = new Triangle(3.0, 4.0, 5.0);
Triangle right = scalene;

isosceles.perimeter();
scalene.perimeter();
right.perimeter();
```

What, if anything, is printed as a result of executing the code segment?

(A)

0.0 12.0

(B)

0.0 12.0 12.0

(C) Nothing is printed because `right` has not been created using the `new` keyword.

(D) Nothing is printed because a `NullPointerException` occurs. ✓

Answer D

Correct. The syntax for calling the `perimeter` method on the `isosceles` object is correct. However, nothing is printed because the `isosceles` variable is assigned the value `null`. Calling a method on a `null` reference results in a `NullPointerException`, causing the execution to terminate.

FINAL Unit1 MCQ - Topic order - Deduplicated

133. Consider the following code segment.

```
String oldStr = "ABCDEF";
String newStr = oldStr.substring(1, 3) + oldStr.substring(4);
System.out.println(newStr);
```

What is printed as a result of executing the code segment?

(A) `ABCD`

(B) `BCDE`

(C) `BCEF` ✓

(D) `BCDEF`

Answer C

Correct. The call `oldStr.substring(1, 3)` returns the substring of `oldStr` starting at index `1` and ending at index `2`, or `"BC"`; the call `oldStr.substring(4)` returns the substring of `oldStr` starting at index `4` and going to the end of `oldStr`, or `"EF"`. Therefore, as the result of the concatenation operation `"+"`, `newStr` gets assigned the value `"BCEF"`.

FINAL Unit1 MCQ - Topic order - Deduplicated

134. Consider the following class definition.

```
public class Beverage
{
  private int temperature;
  public Beverage(int t)
  {
    temperature = t;
  }
  public int getTemperature()
  {
    return temperature;
  }
  public boolean equals(Object other)
  {
    if (other == null)
    {
      return false;
    }
    Beverage b = (Beverage) other;
    return (b.getTemperature() == temperature);
  }
}
```

The following code segment appears in a class other than `Beverage`. Assume that `x` and `y` are properly declared and initialized `int` variables.

```
Beverage hotChocolate = new Beverage(x);
Beverage coffee = new Beverage(y);
boolean same = /* missing code */;
```

Which of the following can be used as a replacement for `/* missing code */` so that the `boolean` variable `same` is set to `true` if and only if the `hotChocolate` and `coffee` objects have the same temperature values?

(A) `(hotChocolate == coffee)`

(B) `hotChocolate.equals(coffee)` ✓

(C) `hotChocolate.equals(coffee.getTemperature())`

(D) `hotChocolate.getTemperature().equals(coffee.getTemperature())`

Answer B

Correct. The correct syntax for the comparison of the objects `hotChocolate` and `coffee` uses one object name with the dot operator and the other object name as the parameter of the `equals` method. The `equals` method

returns `true` if the `temperature` instance variables of both objects are the same and returns `false` otherwise.

135. Consider the following code segment.

```
String temp = "comp";
System.out.print(temp.substring(0) + " "
                 + temp.substring(1) + " "
                 + temp.substring(2) + " "
                 + temp.substring(3));
```

What is printed as a result of executing the code segment?

(A)

c o m p

(B)

comp com co c

(C) ✓

comp omp mp p

(D)

comp comp comp comp

Answer C

Correct. Each of the four `substring` method calls returns a substring beginning with the character at position `n`. Those are `"comp"`, `"omp"`, `"mp"`, and `"p"`, for `n` equal to `0`, `1`, `2`, and `3`, respectively. The print statement concatenates those return values, separated by spaces, and the string `"comp omp mp p"` is printed.

FINAL Unit1 MCQ - Topic order - Deduplicated

136. Consider the following code segment.

```
String str = "CompSci";
System.out.println(str.substring(0, 3));
int num = str.length();
```

What is the value of `num` after the code segment is executed?

(A)

3

(B)

4

(C)

6

(D) ✓

7

Answer D

Correct. In the first code line, `str` is assigned a string of length `7`. In the second code line, calling the `substring` method on `str` does not affect the value of `str`, so in the third code line, `num` is assigned the length of `str`, which is still `7`.

FINAL Unit1 MCQ - Topic order - Deduplicated

137. Consider the following code segment.

```
String str = "0";
str += str + 0 + 8;
System.out.println(str);
```

What is printed as a result of executing the code segment?

(A) `08`

(B) `008`

(C) `0008` ✓

(D) Nothing is printed, because numerical values cannot be added to a `String` object.

Answer C

Correct. The first line of code initializes the string `str` to `"0"`. In the second line of code, the `int` values `0` and `8` are interpreted as strings. The second line of code concatenates `str`, `"0"`, and `"8"` and then appends the result to `str`, setting `str` to `"0008"`.

138. Consider the following code segment.

```
int one = 1;
int two = 2;
String zee = "Z";
System.out.println(one + two + zee);
```

What is printed as a result of executing the code segment?

(A) `12Z`

(B) `3Z` ✓

(C) `12zee`

(D) `3zee`

Answer B

Correct. The `+` operators in the `println` statement are processed from left to right. Since `one` and `two` are `integer` variables, the code `one + two` performs integer addition, producing `3`; since `zee` is a `String` variable, the code `3 + zee` performs string concatenation after first converting the `int` value `3` to a `String` value.

FINAL Unit1 MCQ - Topic order - Deduplicated

139. The following code segment is intended to assign to newWord the result created by removing the first occurrence of "a" from word. Assume that the String variable word has been properly declared and initialized. This code segment works for some, but not all, values of word.

```
int aLoc = word.indexOf("a");
String newWord = word.substring(0, aLoc) + word.substring(aLoc + 1);
```

Which of the following conditions best describes the condition in which this code segment will not work as intended and will result in a runtime error?

(A) "a" is the first character in word.

(B) "a" is the last character in word.

(C) word does not contain the character "a". ✓

(D) word contains only a sequence of multiple "a" characters.

Answer C

Correct. indexOf returns -1 when the target String cannot be found. Using -1 as a parameter in substring will result in a StringIndexOutOfBoundsException error.

FINAL Unit1 MCQ - Topic order - Deduplicated

140. Which of the following code segments can be used to set the value of the string `str` to `"Good morning, sunshine!"`?

I.
```
String str = "Good " + "morning," + " sunshine!";
```

II.
```
String str = "Good";
str += " morning, sunshine!";
```

III.
```
String str = " morning, ";
str = "Good" + str + "sunshine!";
```

(A) I only

(B) II only

(C) II and III only

(D) I, II, and III ✓

Answer D

Correct. All three code segments will produce the desired result. Code segment I concatenates the strings `"Good "`, `"morning,"`, and `" sunshine!"`. Code segment II creates the string variable `str` with initial value `"Good"` and uses the `+=` string concatenation operator to append `" morning, sunshine!"` to it. Code segment III creates the string variable `str` with initial value `" morning, "` and then concatenates `"Good"`, `str`, and `"sunshine!"`.

FINAL Unit1 MCQ - Topic order - Deduplicated

141. Consider the following code segment.

```
String one = "xyz";
String two = one.substring(2) + one.substring(1)
           + one.substring(0);
int pos = two.indexOf("x");
```

What is the value of pos after executing the code segment?

(A)

-1

(B)

2

(C) ✓

3

(D)

5

Answer C

Correct. The call one.substring(2) returns "z" (the substring starting at index 2 and going to the end of the string). The call one.substring(1) returns "yz" (the substring starting at index 1 and going to the end of the string). The call one.substring(0) returns "xyz" (the substring starting at index 0 and going to the end of the string). The results of these calls are concatenated to assign the value "zyzxyz" to the variable two. The call two.indexOf("x") returns 3 because "x" appears at index 3 of "zyzxyz".

FINAL Unit1 MCQ - Topic order - Deduplicated

142. Consider the following code segment, which is intended to create a String that consists of the last character in word and assign it to the variable lastChar.

```
String word = /* initialization not shown */;  // line 1
int len = word.length();                        // line 2
String lastChar = word.substring(len);          // line 3
```

Which of the following best describes why this code segment will not work as intended?

(A) substring needs to be called with two parameters, so the method call in line 3 should be changed to word.substring(len, len + 1).

(B) len is not the correct index of the last character in word, so the method call in line 3 should be changed to word.substring(len - 1). ✓

(C) len is not the correct index of the last character in word, so the method call in line 3 should be changed to word.substring(len + 1).

(D) substring cannot be called with a variable as the parameter, so the method call in line 3 should be changed to word.substring(word.length()).

Answer B

Correct. Since String objects are indexed starting at 0, the last character will be located at index word.length() – 1.

FINAL Unit1 MCQ - Topic order - Deduplicated

143. Consider the code segment below.

```
int a = 1988;
int b = 1990;

String claim = " that the world's athletes " +
          "competed in Olympic Games in ";

String s = "It is " + true + claim + a +
       " but " + false + claim + b + ".";

System.out.println(s);
```

What, if anything, is printed when the code segment is executed?

(A) `It is trueclaim1998 but falseclaim1990.`

(B) `It is true that the world's athletes competed in Olympic Games in a but false that the world's athletes competed in Olympic Games in b.`

(C) `It is true that the world's athletes competed in Olympic Games in 1988 but false that the world's athletes competed in Olympic Games in 1990.` ✓

(D) Nothing is printed because the code segment does not compile.

Answer C

Correct. The + operator concatenates the `String` literals, `boolean` values, String `variables`, and `int` variables in the order that they appear.

FINAL Unit1 MCQ - Topic order - Deduplicated

144. Consider the following code segment.

```
String oldStr = "ABCDEF";
String newStr = /* missing code */;
System.out.println(newStr);
```

Which of the following expressions can be used to replace /* missing code */ so that the code segment prints the string "BCEF"?

(A)

```
oldStr.substring(1, 2) + oldStr.substring(4)
```

(B)

```
oldStr.substring(1, 2) + oldStr.substring(5)
```

(C) ✓

```
oldStr.substring(1, 3) + oldStr.substring(4)
```

(D)

```
oldStr.substring(1, 3) + oldStr.substring(5)
```

Answer C

Correct. The call oldStr.substring(1, 3) returns the substring beginning at index 1 and ending at index 2 (1 less than the parameter 3). The returned substring is "BC". The call oldStr.substring(4) returns the substring beginning at index 4 and going to the end of the string. The returned substring is "EF". The two returned substrings are concatenated to form "BCEF".

FINAL Unit1 MCQ - Topic order - Deduplicated

145. In the following code segment, `mon` and `tue` are properly declared `String` variables.

```
String str1 = mon.substring(0, mon.indexOf(" "));
String str2 = tue.substring(tue.indexOf(" ") + 1);
System.out.println(str1 + " and " + str2);
```

For which of the following values of `mon` and `tue` will the code segment print `"cloudy and foggy"`?

FINAL Unit1 MCQ - Topic order - Deduplicated

(A)

mon	tue
"cloudy today"	"foggy tomorrow"

(B) ✓

mon	tue
"cloudy today"	"tomorrow foggy"

(C)

mon	tue
"today cloudy"	"foggy tomorrow"

(D)

mon	tue
"today cloudy"	"tomorrow foggy"

Answer B

Correct. The first assignment statement uses the `indexOf` method to find the first occurrence of `" "` in `mon`, which occurs at index 6. It then assigns to `str1` the substring of `mon` from index 0 to index 5, `"cloudy"`. The

second assignment statement also uses the `indexOf` method to find the first occurrence of `" "` in `tue`, which occurs at index 8. It then assigns to `str2` the substring from index 9 to the end of the string, `"foggy"`.

146. Consider the following code segment.

```
String original = "DEFGHI";
String result = /* missing code */;
System.out.println(result);
```

Which of the following expressions can be used to replace /* missing code */ so that this code segment prints the string "HIDE"?

(A)

```
original.substring(4) + original.substring(0, 1)
```

(B) ✓

```
original.substring(4) + original.substring(0, 2)
```

(C)

```
original.substring(5) + original.substring(1, 2)
```

(D)

```
original.substring(5) + original.substring(1, 3)
```

Answer B

Correct. The call original.substring(4) returns "HI", which is the substring starting at index 4 and going through the end of the string. The call original.substring(0, 2) returns "DE", which is the substring starting at index 0 and ending at index 2 - 1. Note that substring(int from, int to) goes from index from to index to - 1.

FINAL Unit1 MCQ - Topic order - Deduplicated

147. Consider the `processWords` method. Assume that each of its two parameters is a `String` of length two or more.

```
public void processWords(String word1, String word2)
{
 String str1 = word1.substring(0, 2);
 String str2 = word2.substring(word2.length() - 1);
 String result = str2 + str1;
 System.out.println(result.indexOf(str2));
}
```

Which of the following best describes the value printed when `processWords` is called?

(A) The value `0` is always printed. ✓

(B) The value `1` is always printed.

(C) The value `result.length() - 1` is printed.

(D) A substring containing the last character of `word2` is printed.

Answer A

FINAL Unit1 MCQ - Topic order - Deduplicated

148. Consider the following code segment.

```
String str1 = "LMNOP";
String str2 = str1.substring(3);
str2 += str1.substring(2, 3);
```

What is the value of str2 after executing this code segment?

(A) ✓

"OPN"

(B)

"OPNO"

(C)

"NOPM"

(D)

"NOPMN"

Answer A

Correct. The call str1.substring(3) returns the substring of str1 starting at index 3, or "OP". This value is initially assigned to str2. The call str1.substring(2, 3) returns the substring of str1 starting at index 2 and ending at index 3 - 1, or "N". This value is appended to str2, resulting in "OPN".

FINAL Unit1 MCQ - Topic order - Deduplicated

149. Consider the following code segment.

```
String one = "computer";
String two = "science";
String concat = /* missing code */;
System.out.println(concat);
```

Which of the following expressions can be used to replace `/* missing code */` so that the code segment prints the string `"pun"`?

(A)

```
one.substring(3, 4) + two.substring(4, 4)
```

(B) ✓

```
one.substring(3, 5) + two.substring(4, 5)
```

(C)

```
one.substring(4, 5) + two.substring(5, 5)
```

(D)

```
one.substring(4, 6) + two.substring(5, 6)
```

Answer B

Correct. The call `one.substring(3, 5)` returns `"pu"`, and the call `two.substring(4, 5)` returns `"n"`. The concatenated result is `"pun"`.